

FC7xxx ETHTRCV User Manual

Rev.A0

Revision History

Revision	Date	Changes
A0	2025/1/22	Release for FC7300F8MDQ support

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Chapter 1 ETHTRCV Introduction

1.1 Requirement Tracing

The design of this module follows the specifications of the EthTrcv driver specified in AUTOSAR Classic Platform Release 4.6.0. For detailed requirements, refer to the *AUTOSAR_SWS_EthernetTransceiverDriver*.

1.2 Hardware Summary

The Ethernet Transceiver (EthTrcv) module is available for FC7300F8MDT or FC7300F8MDQ demo board.

Chapter 2 Software Design

2.1 Rejected Requirements

Rejected Requirement 1 **SWS_EthTrcv_00110**

Description	The Ethernet Transceiver driver shall support wake up detection depending on the configuration parameter EthTrcvWakeUpSupport either not at all (ETHTRCV_WAKEUP_NOT_SUPPORTED) or by interrupt (ETHTRCV_WAKEUP_BY_INTERRUPT) or by asynchronous check (ETHTRCV_WAKEUP_BY_ASYNCHRONOUS_CHECK) or by polling (ETHTRCV_WAKEUP_BY_POLLING).
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 **SWS_EthTrcv_00111**

Description	If the EthTrcv is requested with ETH_MODE_DOWN (e.g. low power mode), the Ethernet Transceiver driver shall enable the corresponding ICU channel (see EthTrcvIcuChannelRef) by calling Icu_EnableNotification if configured.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 **SWS_EthTrcv_00112**

Description	If the EthTrcv is requested with ETH_MODE_ACTIVE (e.g. normal mode), the Ethernet Transceiver driver shall disable the corresponding ICU channel (see EthTrcvIcuChannelRef) by calling Icu_DisableNotification if configured.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 **SWS_EthTrcv_00146**

Description	The wake up interrupt handler (if present) shall clear the interrupt and identify the wake up reason and store it.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 **SWS_EthTrcv_00159**

Description	If EthTrcv_RunPortCableDiagnostic is called, EthTrcv has to ensure that the Ethernet hardware (PHY) is in a state to run the cable diagnostic by considering at least the following points: - the corresponding Ethernet transceiver is in state ETH_MODE_ACTIVE - the corresponding Ethernet transceiver is in state ETHTRCV_LINK_STATE_DOWN If all pre conditions are fulfilled to run the cable diagnostic measurement, EthTrcv shall trigger the cable diagnostic measurement and set the state internally to ETHTRCV_CABLEDIAG_PENDING of the affected Ethernet transceiver.
Rejection Reason	Cable Diagnostic not supported yet.

Rejected Requirement 1 **SWS_EthTrcv_00160**

Description	If EthTrcv_GetCableDiagnosticsResult is called, the current state of the cable diagnostic measurement of the affected Ethernet transceiver shall be returned and stored per Ethernet transceiver as internal cable diagnostic state.
Rejection Reason	Cable Diagnostic not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00161	
Description	As long as the cable diagnostic measurement is running (internal cable diagnostic state is ETHTRCV_CABLEDIAG_PENDING), a mode request (indicated by EthTrcv_SetTransceiverMode) and link request (indicated by EthTrcv_TransceiverLinkStateRequest), respectively, shall be stored and not executed.
Rejection Reason	Cable Diagnostic not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00162	
Description	As soon as the cable diagnostic measurement has finished (internal cable diagnostic state is different from ETHTRCV_CABLEDIAG_PENDING), EthTrcv shall execute the last mode request and link request, respectively, of the affected Ethernet transceiver.
Rejection Reason	Cable Diagnostic not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00115	
Description	The function shall check for wake-up reasons and propagate the corresponding wake-up source (see EthTrcvWakeupMap configuration) to the EcuM by calling EcuM_SetWakeupEvent.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00163	
Description	If EthTrcv_SetTransceiverMode() is called and the internal cable diagnostic state is equal to ETHTRCV_CABLEDIAG_PENDING, the Ethernet Transceiver driver shall store the mode request per EthTrcv (TrcvIdx).
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00179	
Description	A new requested EthTrcv mode shall overwrite the last requested EthTrcv mode, except in case the last mode is ETH_MODE_ACTIVE_WITH_WAKEUP_REQUEST and the new mode is ETH_MODE_ACTIVE.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00117	
Description	If the function is called with ETH_MODE_DOWN and EthTrcvWakeupSleepOnDataLineEnabled is set to FALSE or not configured, it shall set the corresponding Ethernet Hardware (PHY) into a mode (e.g. sleep mode) where wakeups can be detected.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00180	
Description	If the function is called with ETH_MODE_DOWN, EthTrcvWakeupSleepOnDataLineEnabled is set to TRUE, the Ethernet Transceiver driver has detected a pending wake-up and a corresponding delay timer with EthTrcvSleepModeExecutionDelay is not running, the Ethernet Transceiver driver shall start a delay timer with EthTrcvSleepModeExecutionDelay and return with E_OK.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00181	
Description	If the function is called with ETH_MODE_DOWN, EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE, the Ethernet Transceiver driver has detected a pending wake-up and a corresponding delay timer with EthTrcvSleepModeExecutionDelay is running, then the Ethernet Transceiver driver shall return with E_OK.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00182	
Description	If the function is called with ETH_MODE_DOWN, EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE and the EthTrcv has not detected a pending wake-up (see SWS_EthTrcv_00180), the Ethernet hardware (PHY) is in another mode than sleep mode and has not already triggered a sleep request, the Ethernet Transceiver driver shall trigger a sleep request according to the specification of OA TC10 (see [23]).
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00118	
Description	If EthTrcv_SetTransceiverMode() is called with parameter ETH_MODE_ACTIVE or ETH_MODE_ACTIVE_WITH_WAKEUP_REQUEST, EthTrcvWakeupSleepOnDatalineEnabled set to FALSE or not configured and the internal cable diagnostic state is different from ETHTRCV_CABLEDIAG_PENDING, the Ethernet Transceiver driver shall: - Check for wake-up reasons when entering the transceiver's active mode; - If no wake-up reason has been detected, the Ethernet transceiver shall send a wake-up symbol on the bus if configured; - Invoke the call-out <EthTrcvWakeUpCallout> function if configured.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00183	
Description	If the function is called with ETH_MODE_ACTIVE, EthTrcvWakeupSleepOnDatalineEnabled is TRUE and the internal cable diagnostic state is different from ETHTRCV_CABLEDIAG_PENDING, the Ethernet Transceiver driver shall: - Put the Ethernet hardware (PHY) to normal mode, if not already switched automatically by the statemachine of the Ethernet hardware (PHY), - Invoke the call-out <EthTrcvWakeUpCallout> function if configured.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00184	
Description	If the function is called with ETH_MODE_ACTIVE_WITH_WAKEUP_REQUEST, EthTrcvWakeupSleepOnDatalineEnabled is TRUE, the internal cable diagnostic state is different from ETHTRCV_CABLEDIAG_PENDING and EthTrcv does not currently perform wakeup repetition algorithm (see SWS_EthTrcv_00196), the Ethernet Transceiver driver shall: - trigger the Ethernet hardware (PHY) to transmit a wake-up on the network according to the specification of OA TC10 (see [23]), - invoke the call-out <EthTrcvWakeUpCallout> function if configured.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00046	
Description	The function shall be pre compile time configurable On/Off by the configuration parameter: EthTrcvSetTransceiverModeApi.
Rejection Reason	The requirement is obsolete.

Rejected Requirement 1 SWS_EthTrcv_00053	
Description	The function shall be pre compile time configurable On/Off by the configuration parameter: EthTrcvGetTransceiverModeApi.
Rejection Reason	The requirement is obsolete.

Rejected Requirement 1 SWS_EthTrcv_00119	
Description	Service Name EthTrcv_SetTransceiverWakeupMode (obsolete) Std_ReturnType EthTrcv_SetTransceiverWakeupMode (uint8 TrcvIdx, Syntax EthTrcv_WakeupModeType TrcvWakeupMode) Service ID 0x0d [hex] Sync/Async Synchronous Reentrancy Non Reentrant Parameters (in) TrcvIdx Index of the transceiver within the context of the Ethernet Transceiver Driver TrcvWakeupMode ETHTRCV_WUM_DISABLE: disable transceiver wake up ETHTRCV_WUM_ENABLE: enable transceiver wake up ETHTRCV_WUM_CLEAR: clears transceiver wake up reason Parameters (inout) None Parameters (out) None Return value Std_ReturnType E_OK: transceiver wake up mode has been changed. E_NOT_OK: transceiver wake up mode could not be changed or the wake-up reason could not be cleared. Description Enables / disables the wake-up mode or clear the wake-up reason of the indexed transceiver Tags:atp.Status=obsolete Available via EthTrcv.h
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00128	
Description	The function EthTrcv_SetTransceiverWakeupMode() shall read the current transceiver wake up mode and provide it into TrcvWakeupModePtr.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00129	
Description	If development error detection is enabled: The function EthTrcv_SetTransceiverWakeupMode() shall check that the service EthTrcv_Init was previously called. If the check fails, the function shall raise the development error ETHTRCV_E_UNINIT otherwise (if DET is disabled) return E_NOT_OK.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00130

Description	If development error detection is enabled: The function EthTrcv_SetTransceiverWakeupMode() shall check the parameter TrcvIdx for being valid. If the check fails, the function shall raise the development error ETHTRCV_E_INV_TRCV_IDX otherwise (if DET is disabled) return E_NOT_OK.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00131

Description	If development error detection is enabled: The function EthTrcv_SetTransceiverWakeupMode() shall check the parameter TrcvWakeupModePtr for being valid. If the check fails, the function shall raise the development error ETHTRCV_E_PARAM_POINTER otherwise (if DET is disabled) return E_NOT_OK.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00132

Description	The function EthTrcv_SetTransceiverWakeupMode() shall be only available if EthTrcvSetTransceiverWakeupModeApi is set to TRUE.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00133

Description	Caveat: The function EthTrcv_GetTransceiverWakeupMode() requires previous transceiver initialization (EthTrcv_Init).
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00127

Description	Service Name EthTrcv_GetTransceiverWakeupMode (obsolete) Syntax <pre>Std_ReturnType EthTrcv_GetTransceiverWakeupMode (uint8 TrcvIdx, EthTrcv_WakeupModeType* TrcvWakeupModePtr)</pre> Service ID[hex] 0x0e Sync/Async Synchronous Reentrancy Non Reentrant Parameters (in) TrcvIdx Index of the transceiver within the context of the Ethernet Transceiver Driver Parameters (inout) None Parameters (out) TrcvWakeupModePtr ETHTRCV_WUM_DISABLE: transceiver wake up is disabled ETHTRCV_WUM_ENABLE: transceiver wake up is enabled Return value Std_ReturnType - E_OK: success - E_NOT_OK: transceiver wake up mode could not be obtained Description Returns the wake up mode of the indexed transceiver. Tags:atp.Status=obsolete Available via EthTrcv.h
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00120

Description	If function EthTrcv_SetTransceiverWakeupMode() is called with ETHTRCV_WUM_DISABLE or ETHTRCV_WUM_ENABLE and the internal cable diagnostic state is different from ETHTRCV_CABLEDIAG_PENDING, the Ethernet Transceiver shall put the indexed transceiver in the specified wake up mode.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00121

Description	If function EthTrcv_SetTransceiverWakeupMode() is called with ETHTRCV_WUM_CLEAR and the internal cable diagnostic state is different from ETHTRCV_CABLEDIAG_PENDING, the Ethernet Transceiver driver shall clear stored wakeup events on the indexed transceiver.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00164

Description	If the internal cable diagnostic state is ETHTRCV_CABLEDIAG_PENDING, the EthTrcv_SetTransceiverWakeupMode shall return E_NOT_OK.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00122

Description	If development error detection is enabled: The function EthTrcv_SetTransceiverWakeupMode() shall check that the service EthTrcv_Init was previously called. If the check fails, the function shall raise the development error ETHTRCV_E_UNINIT otherwise (if DET is disabled) return E_NOT_OK.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00123

Description	If development error detection is enabled: The function EthTrcv_SetTransceiverWakeupMode() shall check the parameter TrcvIdx for being valid. If the check fails, the function shall raise the development error ETHTRCV_E_INV_TRCV_IDX otherwise (if DET is disabled) return E_NOT_OK.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00124

Description	The function EthTrcv_GetTransceiverWakeupMode() shall be only available if EthTrcvWakeUpSupport is not disabled (set to ETHTRCV_WAKEUP_NOT_SUPPORTED).
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00125

Description	If the transceiver is already in the requested wake-up mode, E_OK shall be returned and no development error shall be raised.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00126

Description	Caveat: The function EthTrcv_GetTransceiverWakeupMode() requires previous transceiver initialization (EthTrcv_Init).
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Rejection Reason	Wakeup not supported yet.
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Rejected Requirement 1 SWS_EthTrcv_91012	
Description	Service Name EthTrcv_EthTrcv_GetBusWuReason (draft) Syntax Std_ReturnType EthTrcv_EthTrcv_GetBusWuReason (uint8 TrcvIdx, EthTrcv_WakeupReasonType* WakeupReasonPtr) Service ID [hex] 0x17 Sync/Async Synchronous Reentrancy Reentrant Parameters (in) TrcvIdx Index of the transceiver within the context of the Ethernet Transceiver Driver Parameters (inout) None Parameters (out) WakeupReasonPtr Pointer to structure of least recent wakeup event, which was detected by the Ethernet PHY Return value Std_ReturnType E_OK: PHY wake up reason request has been accepted. E_NOT_OK: PHY wake up reason request has not been accepted. Description This function returns the least recent wakeup reasons. Tags:atp.Status=draft Available via EthTrcv.h
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00186	
Description	The function EthTrcv_GetBusWuReason shall read the stored wake-up reason and provide the information in WakeupReasonPtr.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00187	
Description	If development error detection is enabled: the function shall check that the service EthTrcv_Init was previously called. If the check fails, the function shall raise the development error ETHTRCV_E_UNINIT otherwise (if DET is disabled) return E_NOT_OK.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00188	
Description	If development error detection is enabled: the function shall check the parameter TrcvIdx for being valid. If the check fails, the function shall raise the development error ETHTRCV_E_INV_TRCV_IDX otherwise (if DET is disabled) return E_NOT_OK.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00189	
Description	If development error detection is enabled: the function shall check the parameter TrcvModePtr for being valid. If the check fails, the function shall raise the development error ETHTRCV_E_PARAM_POINTER otherwise (if DET is disabled) return E_NOT_OK.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00190	
Description	The function shall be pre compile time configurable On/Off by the configuration parameter: EthTrcvGetBusWuReasonApi.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00134	
Description	Service Name EthTrcv_CheckWakeup Syntax Std_ReturnType EthTrcv_CheckWakeup (uint8 TrcvIdx) Service ID [hex] 0x0f Sync/Async Synchronous Reentrancy Reentrant Parameters (in) TrcvIdx Index of the transceiver within the context of the Ethernet Transceiver Driver Parameters (inout) None Parameters (out) None Return value Std_ReturnType E_NOT_OK: The function could not be successfully executed E_OK: The function has been successfully executed Description Service is called by EthIf in case a wake-up interrupt is detected. Available via EthTrcv.h
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00136	
Description	If the wake-up mode of the corresponding transceiver is not ETHTRCV_WUM_ENABLE, the function EthTrcv_CheckWakeup() shall return E_OK.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00135	
Description	If EthTrcvWakeUpSupport is set to ETHTRCV_WAKEUP_BY_INTERRUPT, the function EthTrcv_CheckWakeup() shall check if a wake up has been detected and if yes propagate the corresponding wake up source (see EthTrcvWakeupMap configuration) to the EcuM by calling EcuM_SetWakeupEvent.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00185	
Description	If the function EthTrcv_CheckWakeup() is called and EthTrcvWakeUpSupport is set to ETHTRCV_WAKEUP_BY_ASYNCHRONOUS_CHECK, the Ethernet Transceiver Driver shall store the request to asynchronously check for a wake-up of used Ethernet hardware (e.g. the OA TC10 compliant PHY) in the context of the EthTrcv_MainFunction.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00137	
Description	If development error detection is enabled: The function EthTrcv_CheckWakeup() shall check that the service EthTrcv_Init was previously called. If the check fails, the function shall raise the development error ETHTRCV_E_UNINIT otherwise (if DET is disabled) return E_NOT_OK.

Rejection Reason	Wakeup not supported yet.
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Rejected Requirement 1 SWS_EthTrcv_00138

Description	If development error detection is enabled: The function EthTrcv_CheckWakeup() shall check the parameter TrcvIdx for being valid. If the check fails, the function shall raise the development error ETHTRCV_E_INV_TRCV_IDX otherwise (if DET is disabled) return E_NOT_OK.
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Rejection Reason	Wakeup not supported yet.
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Rejected Requirement 1 SWS_EthTrcv_00139

Description	The function EthTrcv_CheckWakeup() shall be only available if EthTrcvWakeUpSupport is something else than ETHTRCV_WAKEUP_NOT_SUPPORTED.
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Rejection Reason	Wakeup not supported yet.
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Rejected Requirement 1 SWS_EthTrcv_00140

Description	Caveat: The function EthTrcv_CheckWakeup() requires previous transceiver initialization (EthTrcv_Init).
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Rejection Reason	Wakeup not supported yet.
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Rejected Requirement 1 SWS_EthTrcv_91025

Description	Service Name EthTrcv_TransceiverLinkStateRequest Syntax Std_ReturnType EthTrcv_TransceiverLinkStateRequest (uint8 TrcvIdx, EthTrcv_LinkStateType LinkState) Service ID [hex] 0x02 Sync/Async Asynchronous Reentrancy Reentrant for different TrcvIdx. Non reentrant for the same TrcvIdx. Parameters (in) TrcvIdx Index of the transceiver within the context of the Ethernet Transceiver Driver LinkState The Ethernet link state of a physical Ethernet connection. Parameters (inout) None Parameters (out) None Return value Std_ReturnType E_OK: The request has been accepted E_NOT_OK: The request has not been accepted Description Request the given link state for the given Ethernet transceiver Available via EthTrcv.h
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Rejection Reason	Link State Request not supported yet.
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Rejected Requirement 1 SWS_EthTrcv_00151

Description	The function shall start link training of the indexed transceiver if: - the requested link state is ETHTRCV_LINK_STATE_ACTIVE and - the internal cable diagnostic state is different from ETHTRCV_CABLEDIAG_PENDING and - EthTrcvConnNeg is set to TRCV_CONN_NEG_MASTER or TRCV_CONN_NEG_AUTO. If EthTrcvConnNeg is set to TRCV_CONN_NEG_SLAVE, the indexed transceiver shall be put in a state to wait for the link training of the link partner.
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Rejection Reason	Link State Request not supported yet.
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Rejected Requirement 1 SWS_EthTrcv_00165	
Description	If EthTrcv_TransceiverLinkStateRequest() is called and the internal cable diagnostic state is ETHTRCV_CABLEDIAG_PENDING, the Ethernet Transceiver driver shall store the link state request per Ethernet transceiver and proceed as specified in SWS_EthTrcv_00162.
Rejection Reason	Link State Request not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00152	
Description	The function shall stop link training of the indexed transceiver, if the requested link state is ETHTRCV_LINK_STATE_DOWN and EthTrcvConnNeg is set to TRCV_CONN_NEG_MASTER or TRCV_CONN_NEG_AUTO.
Rejection Reason	Link State Request not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00153	
Description	The function shall put the link down of the indexed transceiver, if the requested link state is ETHTRCV_LINK_STATE_DOWN.
Rejection Reason	Link State Request not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00154	
Description	If the Ethernet transceiver is already in the requested link state, E_OK shall be returned and no development error shall be raised.
Rejection Reason	Link State Request not supported yet.

Rejected Requirement 1 SWS_EthTrcv_91003	
Description	Service Name EthTrcv_SetPhyTestMode Syntax Std_ReturnType EthTrcv_SetPhyTestMode (uint8 TrcvIdx, EthTrcv_PhyTestModeType Mode) Service ID [hex] 0x11 Sync/Async Synchronous Reentrancy Reentrant for different TrcvIdx. Non reentrant for the same TrcvIdx. Parameters (in) TrcvIdx Index of the transceiver within the context of the Ethernet Transceiver Driver Mode Test mode to be activated Parameters (inout) None Parameters (out) None Return value Std_ReturnType E_OK: The request has been accepted E_NOT_OK: The request has not been accepted. Description Activates a given test mode. Available via EthTrcv.h
Rejection Reason	Set Phy Test Mode not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00166	
Description	If the internal cable diagnostic state is ETHTRCV_CABLEDIAG_PENDING, the EthTrcv_SetPhyTestMode shall return E_NOT_OK.

Rejection Reason	Set Phy Test Mode not supported yet.
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Rejected Requirement 1 SWS_EthTrcv_00147	
Description	If development error detection is enabled: the function EthTrcv_SetPhyTestMode shall check the parameter Mode for being supported by the hardware. If the check fails, the function shall raise the development error ETHTRCV_E_NOT_SUPPORTED.
Rejection Reason	Set Phy Test Mode not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00169	
Description	The function shall be pre compile time configurable On/Off by the configuration parameter EthTrcvSetPhyTestModeApi (ECUC_EthTrcv_00047).
Rejection Reason	Set Phy Test Mode not supported yet.

Rejected Requirement 1 SWS_EthTrcv_91001	
Description	Service Name EthTrcv_GetPhySignalQuality Syntax Std_ReturnType EthTrcv_GetPhySignalQuality (uint8 TrcvIdx, uint32* SignalQualityPtr) Service ID [hex] 0x10 Sync/Async Synchronous Reentrancy Reentrant for different TrcvIdx. Non reentrant for the same TrcvIdx. Parameters (in) TrcvIdx Index of the transceiver within the context of the Ethernet Transceiver Driver Parameters (inout) None Parameters (out) SignalQualityPtr Pointer to the memory where the signal quality shall be stored. Return value Std_ReturnType E_OK: The request has been accepted E_NOT_OK: The request has not been accepted. Description Obtains the current signal quality of the link of the indexed transceiver Available via EthTrcv.h
Rejection Reason	Get Phy Signal Quality not supported yet.

Rejected Requirement 1 SWS_EthTrcv_91007	
Description	Service Name EthTrcv_SetPhyTxMode Std_ReturnType EthTrcv_SetPhyTxMode (uint8 TrcvIdx, Syntax EthTrcv_PhyTxModeType Mode) Service ID [hex] 0x13 Sync/Async Synchronous Reentrancy Reentrant for different TrcvIdx. Non reentrant for the same TrcvIdx. Index of the transceiver within the context of the Ethernet TrcvIdx Transceiver Driver Parameters (in) Mode Transmission mode to be activated AUTOSAR CP R20-11 Parameters None (inout) Parameters (out) None Std_Return- E_OK: The request has been accepted Return value Type E_NOT_OK: The request has not been accepted Description Activates a given transmission mode. Available via EthTrcv.h
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00168	
Description	If the internal cable diagnostic state is ETHTRCV_CABLEDIAG_PENDING, the EthTrcv_SetPhyTxMode shall return E_NOT_OK.

Rejection Reason	Wakeup not supported yet.
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Rejected Requirement 1 SWS_EthTrcv_00148	
Description	If development error detection is enabled: the function EthTrcv_SetPhyTxMode shall check the parameter Mode for being supported by the hardware. If the check fails, the function shall raise the development error ETHTRCV_E_NOT_SUPPORTED.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_91011	
Description	Service Name EthTrcv_RunCableDiagnostic Syntax Std_ReturnType EthTrcv_RunCableDiagnostic (uint8 TrcvIdx) Service ID [hex] 0x16 Sync/Async Asynchronous Reentrancy Reentrant for different TrcvIdx. Non reentrant for the same TrcvIdx. Parameters (in) TrcvIdx Index of the Ethernet transceiver within the context of the Ethernet Transceiver Driver. Parameters (inout) None Parameters (out) None Return value Std_ReturnType E_OK: The trigger has been accepted. E_NOT_OK: The trigger has not been accepted. Description Trigger the cable diagnostics for the given Ethernet transceiver. Available via EthTrcv.h
Rejection Reason	Run Cable Diagnostic not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00170	
Description	The function shall be pre compile time configurable On/Off by the configuration parameter EthTrcvEnableCableDiagnosticApi (ECUC_EthTrcv_00054).
Rejection Reason	Run Cable Diagnostic not supported yet.

Rejected Requirement 1 SWS_EthTrcv_91009	
Description	Service Name EthTrcv_GetCableDiagnosticsResult Syntax Std_ReturnType EthTrcv_GetCableDiagnosticsResult (uint8 TrcvIdx, EthTrcv_CableDiagResultType* ResultPtr) Service ID [hex] 0x14 Sync/Async Synchronous Reentrancy Reentrant for different TrcvIdx. Non reentrant for the same TrcvIdx. Parameters (in) TrcvIdx Index of the transceiver within the context of the Ethernet Transceiver Driver Parameters (inout) None Parameters (out) ResultPtr Pointer to the location where the cable diagnostics result shall be stored Return value Std_ReturnType E_OK: The request has been accepted

	E_NOT_OK: The request has not been accepted Description Retrieves the cable diagnostics result of a given transceiver. Available via EthTrcv.h
Rejection Reason	Get Cable Diagnostics Result not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00171

Description	The function shall be pre compile time configurable On/Off by the configuration parameter EthTrcvEnableCableDiagnosticApi (ECUC_EthTrcv_00054).
Rejection Reason	Get Cable Diagnostics Result not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00191

Description	If EthTrcvWakeUpSupport is set to ETHTRCV_WAKEUP_BY_ASYNCHRONOUS_CHECK, the EthTrcv_MainFunction shall check the maintained Ethernet hardware (PHY) for a signaled Wakeup.Indication of those EthTrcv's where the check for wake-up was requested by EthTrcv_CheckWakeup. If a wake-up is detected, the Ethernet Transceiver Driver shall inform the EcuM by calling EcuM_SetWakeupEvent with the corresponding WakeupSource.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00192

Description	If EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE and EthTrcvActAsSlavePassiveEnabled is set to TRUE, the EthTrcv_MainFunction shall evaluate the maintained Ethernet hardware (PHY) for a signaled Sleep.Indication (see [23]). If a Sleep.Indication is detected, the Ethernet Transceiver Driver shall inform the EthIf by calling EthIf_SleepIndication.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00193

Description	If EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE, the EthTrcv_MainFunction shall process running delay timers. If a sleep delay timer exceeds and the requested transceiver mode is still ETH_MODE_DOWN, the Ethernet Transceiver driver shall trigger a sleep request according to the specification of OA TC10 (see [23]).
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00194

Description	If EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE, the requested transceiver mode is still ETH_MODE_DOWN and SleepFail.Indication was reported less than EthTrcvSleepRequestNumberOfRepetitions, the Ethernet Transceiver driver shall trigger a sleep request according to the specification of OA TC10 (see [23]) after EthTrcvSleepRequestRepetitionPeriod has been elapsed.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00195

Description	If EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE, the requested transceiver mode is still ETH_MODE_DOWN and SleepFail.Indication was reported EthTrcvSleepRequestNumberOfRepetitions times, the Ethernet Transceiver driver shall call
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	EthIf_TrcvModeIndication(ETH_MODE_DOWN). If EthTrcvForceSleepEnabled is set to true the Ethernet Transceiver driver shall additionally force the Ethernet hardware (PHY) to go to sleep.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00196

Description	After each call of EthTrcv_SetTransceiverMode with ETH_MODE_ACTIVE_WITH_WAKEUP_REQUEST for a particular EthTrcv the EthTrcv_MainFunction shall repeat a wake-up request EthTrcvWakeupRequestNumberOfRepetitions times with a delay of EthTrcvWakeupRequestRepetitionPeriod in between as long as no EthTrcv_SetTransceiverMode with ETH_MODE_DOWN occurs for the same EthTrcv.
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00144

Description	Service <EthTrcvWakeUpCallout> (draft) Name void <EthTrcvWakeUpCallout> (uint8 TrcvIdx, Syntax Eth_ModeType TrcvMode) Service ID 0x11 [hex] Sync/Async Asynchronous Reentrancy Non Reentrant Dont care Trcv Index of the Ethernet Transceiver Idx Parameters ETH_MODE_ACTIVE: enable the transceiver ETH_MODE_ACTIVE_ (in) Trcv WITH_WAKEUP_REQUEST: enable the transceiver and request to trigger Mode a wake-up on the network, if the used Ethernet PHY support such a feature. E.g. used for PHYs with a compliant to OA TC10. Parameters None (inout) Parameters None (out) Return value None Description Indicates an wake-up request for the specified Ethernet Transceiver. The given Trcv AUTOSAR CP R20-11 Mode indicates if the request is based on a remote request from the network (ETH_MODE_ACTIVE) or if the request is based on active user request (ETH_MODE_ACTIVE_WITH_WAKEUP_REQUEST). The callback function Ccan be used to trigger integrator code that initiates a remote wake-up .(e.g. trigger a wake-up line for an active user request). Tags:atp.Status=draft Available via EthTrcv_Externals.h
Rejection Reason	Wakeup not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00071

Description	Name: EthTrcvActAsSlavePassiveEnabled Parent Container: EthTrcvConfig Description: Specifies if the ECU is acting as a passive communication slave on the corresponding ComM channel (corresponding ComM channel has ComMNMVariant set to SLAVE_PASSIVE). If the parameter is set to TRUE, the Ethernet transceiver driver shall poll the maintained Ethernet hardware for a signaled Sleep.Indication (according to OA TC10) in the context of the EthTrcv_MainFunction.Tags:atp.Status=draft Multiplicity: 0..1 Type: EcucBooleanParamDef Default value: -- Post-Build Variant Multiplicity: false Post-Build Variant Value: false Multiplicity Configuration Class: Pre-compile time X All Variants Link time -- Post-build time --
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	Value Configuration Class: Pre-compile time X All Variants Link time -- Post-build time -- scope: local dependency: This parameter is only valid, if EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE and the corresponding ComM channel has set ComMNMVariant to SLAVE_PASSIVE.
Rejection Reason	This parameter is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00074

Description	Name: EthTrcvForceSleepEnabled Parent Container: EthTrcvConfig Description: Specifies if the Ethernet hardware (PHY) support to go to sleep without the confirmation of the Ethernet hardware (PHY) of the linked Ethernet communication partner according to the OA TC10. If the parameter is set to TRUE, the Ethernet hardware (PHY) support to force its hardware state to the sleep state. This is used, if the Ethernet communication partner do not accept the Sleep.Request after the specified repetitions on Sleep.Request (see EthTrcvSleepRequestNumberOfRepetitions). If EthTrcvSleepRequestNumberOfRepetitions is not available, the EthTransceiver driver shall force the Ethernet hardware (PHY) to sleep state after the first unsuccessful Sleep.Request. Tags: atp.Status=draft Multiplicity: 1 Type: EcucBooleanParamDef Default value: -- Post-Build Variant Multiplicity: false Post-Build Variant Value: false Value Configuration Class: Pre-compile time X All Variants Link time -- Post-build time -- scope: local dependency: This parameter is only valid, if EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE.
Rejection Reason	This parameter is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00056

Description	Name EthTrcvPhysLayerPlcaLocalNodeId Parent Container EthTrcvConfig Description Configuration parameter for the transceiver node ID when the PLCA mode for 10BASE-T1S is used. Tags: atp.Status=draft Multiplicity 0..1 Type EcucIntegerParamDef Range 0 .. 255 Default value 255 Post-Build Variant Value false
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	Multiplicity Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local dependency: This parameter is only relevant in case that EthTrcvEnablePLCA is set to true. If the value is set to 0, the transceiver is configured as head-node. Note: Within a PLCA cluster the head-node ID shall be unique. A headnode schedules the transmission cycles and the node ID at the client level defines the transmission order.
Rejection Reason	PLCA(Physical Layer Collision Avoidance) is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00059

Description	Name EthTrcvPhysLayerPlcaMaxBurstCount Parent Container EthTrcvConfig Description Defines maximum packets allowed to be transmitted within a TO. The maximum burst count could differ per ECU within a PLCA mixed segment. Tags: atp.Status=draft Multiplicity 0..1 Type EcucIntegerParamDef Range 0 .. 255 Default value -- Post-Build Variant Value false Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local dependency: This parameter is only relevant in case that EthTrcvEnablePLCA is set to true. If EthTrcvPhysLayerPlcaMaxBurstCount is set to 0 (default value), only one ethernet frame per TO shall be transmitted.
Rejection Reason	PLCA(Physical Layer Collision Avoidance) is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00060

Description	Name EthTrcvPhysLayerPlcaMaxBurstTimer Parent Container EthTrcvConfig Description Limits the burst frames in bit time. The maximum burst time could differ per ECU within a PLCA mixed segment. Tags: atp.Status=draft Multiplicity 0..1 Type EcucIntegerParamDef Range 0 .. 255 Default value 128 Post-Build Variant Value false
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	Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local dependency: This parameter is only relevant in case that EthTrcvEnablePLCA is set to true. Note: For PLCA burst mode to work properly this time should be set greater than one IPG.
Rejection Reason	PLCA(Physical Layer Collision Avoidance) is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00058

Description	Name EthTrcvPhysLayerPlcaNodeCount Parent Container EthTrcvConfig Description Defines the number of communication participants on the mixed segment. This value is relevant for the head-node transceiver (EthTrcvPhysLayerNodeID = 0) in order to know when the BEACON has to be transmitted. Tags: atp.Status=draft Multiplicity 0..1 Type EcuIntegerParamDef Range 0 .. 255 Default value 8 Post-Build Variant Value false Multiplicity Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local dependency: This parameter is only relevant in case that EthTrcvEnablePLCA is set to true.
Rejection Reason	PLCA(Physical Layer Collision Avoidance) is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00057

Description	Name EthTrcvPhysLayerPlcaTransmitOpportunityTimer Parent Container EthTrcvConfig Description Timer for the transmission in bit time to evaluate if a Transmission Opportunity is yield or not. Tags: atp.Status=draft Multiplicity 0..1 Type EcuIntegerParamDef Range 1 .. 255 Default value 32 Post-Build Variant Value false Multiplicity Configuration Class Pre-compile time X All Variants
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	Link time -- Post-build time -- Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local dependency: This parameter is only relevant in case that EthTrcvEnablePLCA is set to true. This parameter shall be set identical within a PLCA cluster.
Rejection Reason	PLCA(Physical Layer Collision Avoidance) is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00063

Description	Name EthTrcvSleepModeExecutionDelay Parent Container EthTrcvConfig Description Specifies the time delay in seconds to execute a sleep (see OA TC10) for a Ethernet hardware (PHY), if a pending wake-up was detected while a ETH_MODE_DOWN was requested. The value shall be an integral multiple of EthTrcvMainFunctionPeriod. Tags:atp.Status=draft Multiplicity 0..1 Type EcucFloatParamDef Range [0 .. INF] Default value -- Post-Build Variant false Multiplicity Post-Build Variant Value false Multiplicity Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local dependency: This parameter is only valid, if EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE.
Rejection Reason	Wakeup is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00072

Description	Name EthTrcvSleepRequestNumberOfRepetitions Parent Container EthTrcvConfig Description Specifies the repetitions to trigger a Sleep.Request (according to OA TC10) if a release of the communication channel was triggered by the upper layer (ETH_MODE_DOWN) and a SleepFail.Indication was signaled. Thus, the Ethernet hardware (PHY) of the connected communication partner did not accept the Sleep.Request. Tags: atp.Status=draft
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	Multiplicity 1 Type EcucIntegerParamDef Range 0 .. 255 Default value 0 Post-Build Variant Value false Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local dependency: This parameter is only valid, if EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE.
Rejection Reason	This parameter is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00073

Description	Name EthTrcvSleepRequestRepetitionPeriod Parent Container EthTrcvConfig Description Specifies the repetition period in seconds of repetitions for a Sleep Request (according to OA TC10). The value shall be a integral multiple of EthTrcvMainFunctionPeriod. Tags: atp.Status=draft Multiplicity 1 Type EcucFloatParamDef Range]0 .. INF[Default value -- Post-Build Variant Value false Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local dependency: This parameter is only valid, if EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE and EthTrcvSleepRequestNumberOfRepetitions is set greater than 0.
Rejection Reason	This parameter is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00028

Description	Name EthTrcvWakeUpCallout Parent Container EthTrcvConfig Description Configuration of the call-out name. Multiplicity 0..1 Type EcucFunctionNameDef Default value -- maxLength -- minLength -- regularExpression --
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	Post-Build Variant Multiplicity false Post-Build Variant Value false Multiplicity Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency dependency: Only valid if EthTrcvWakeUpSupport is not ETHTRCV_WAKEUP_NOT_SUPPORTED.
Rejection Reason	This parameter is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00067

Description	Name EthTrcvWakeupForwardLocalEnabled Parent Container EthTrcvConfig Description Specifies if remote wake up forwarding is enabled (TRUE) or disabled (FALSE) for OA TC10 compliant Ethernet Transceiver.If the parameter is set to TRUE, the Ethernet hardware (PHY) activate a local wake up (e.g. via I/O pin), if a remote wake-up on data line (e.g.100Base-T1) was received (either WUP or WUR). Tags:atp.Status=draft Multiplicity 1 Type EcucBooleanParamDef Default value -- Post-Build Variant Value false Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local dependency: This parameter is only valid, if EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE and if EthTrcvWakeupLocalEnabled is set to TRUE.
Rejection Reason	Wake up is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00068

Description	Name EthTrcvWakeupForwardRemoteEnabled Parent Container EthTrcvConfig Description Specifies if local wake up forwarding is enabled (TRUE) or disabled (FALSE) for OA TC10 compliant Ethernet Transceiver. If the parameter is set to TRUE, the Ethernet hardware (PHY) transmit a wake-up (WUP or WUR on the data line (e.g. 100Base-T1), when a local wake-up occurred (e.g. via I/O pin), which was triggered by neighboring Ethernet hardware (PHY). Tags:
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	atp.Status=draft Multiplicity1 Type EcucBooleanParamDef Default value -- Post-Build Variant Value false Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local dependency: This parameter is only valid, if EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE and if EthTrcvWakeupRemoteEnabled set to TRUE.
Rejection Reason	Wake up is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00069

Description	Name EthTrcvWakeupLocalDetectionTime Parent Container EthTrcvConfig Description Defines the time in seconds when a local wake-up (e.g. via I/O pin) triggered by a neighboring PHY is evalutated as a valid wake-up.Tags: atp.Status=draft Multiplicity 0..1 Type EcucFloatParamDef Range [1E-6 .. 0.1] Default value -- Post-Build Variant Multiplicity false Post-Build Variant Value false Multiplicity Configuration Class Pre-compile time X VARIANT-PRE-COMPILE Link time X VARIANT-LINK-TIME, VARIANT-POST-BUILD Post-build time -- Value Configuration Class Pre-compile time X VARIANT-PRE-COMPILE Link time X VARIANT-LINK-TIME, VARIANT-POST-BUILD Post-build time -- Scope / Dependency scope: local dependency: This parameter is only valid, ifEthTrcvWakeupSleepOnDatalineEnabled is set to TRUE.
Rejection Reason	Wake up is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00070

Description	Name EthTrcvWakeupLocalDurationTime Parent Container EthTrcvConfig Description Defines the duration time in seconds how long a local wake-up should be present on the local wake-up connection (e.g. via I/O pin) to indicate the neighboring PHYs about a wake-up. Tags:atp.Status=draft
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	Multiplicity 0..1 Type EcucFloatParamDef Range [1E-6 .. 0.1] Default value -- Post-Build Variant Multiplicity false Post-Build Variant Value false Multiplicity Configuration Class Pre-compile time X VARIANT-PRE-COMPILE Link time X VARIANT-LINK-TIME, VARIANT-POST-BUILD Post-build time -- Value Configuration Class Pre-compile time X VARIANT-PRE-COMPILE Link time X VARIANT-LINK-TIME, VARIANT-POST-BUILD Post-build time -- Scope / Dependency scope: local dependency: This parameter is only valid, if EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE.
Rejection Reason	Wake up is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00065

Description	Name EthTrcvWakeupLocalEnabled Parent Container EthTrcvConfig Description Specifies if local wake-up is enabled (TRUE) or disabled (FALSE) for OATC10 compliant Ethernet hardware (PHY).If the parameter is set to TRUE, the Ethernet hardware (PHY) is able to detect and react on a wake-up received by a neighboring PHY (e.g. via I/O pin). Tags:atp.Status=draft Multiplicity 1 Type EcucBooleanParamDef Default value -- Post-Build Variant Value false Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local dependency: This parameter is only valid, if EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE.
Rejection Reason	Wake up is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00066

Description	Name EthTrcvWakeupRemoteEnabled Parent Container EthTrcvConfig Description Specifies if remote wake-up is enabled (TRUE) or disabled (FALSE) for OA TC10 compliant Ethernet hardware (PHY).If the parameter is set to TRUE, the Ethernet hardware (PHY) wake up when receiving a remote wake-up (e.g. via 100Base-T1 data line) triggered by
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	the connected communication partner. Tags:atp.Status=draft Multiplicity 1 Type EcucBooleanParamDef Default value -- Post-Build Variant Value false Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local dependency: This parameter is only valid, if EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE.
Rejection Reason	Wake up is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00075

Description	Name EthTrcvWakeupRequestNumberOfRepetitions Parent Container EthTrcvConfig Description Specifies the repetitions to trigger a wake-up request (according to OA TC10), if an active communication request (ETH_MODE_ACTIVE_WITH_WAKEUP_REQUEST) was triggered by the upperlayer. Tags:atp.Status=draft Multiplicity 1 Type EcucIntegerParamDef Range 0 .. 255 Default value 0 Post-Build Variant Value false Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local dependency: This parameter is only valid, if EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE.
Rejection Reason	Wake up is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00062

Description	Name EthTrcvWakeUpRequestRepetitionPeriod Parent Container EthTrcvConfig Description Specifies the repetition period in seconds of a wake-up request, if an active communication request (ETH_MODE_ACTIVE_WITH_WAKEUP_REQUEST) was triggered by the upperlayer. The value shall be an integral multiple of EthTrcvMainFunctionPeriod. Tags: atp.Status=draft Multiplicity 0..1
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	Type EcucFloatParamDef Range]0 .. INF[Default value -- Post-Build Variant Multiplicity false Post-Build Variant Value false Multiplicity Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local dependency: This parameter is only valid, if EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE and EthTrcvSleepRequestNumberOfRepetitions is set greater than 0.
Rejection Reason	Wake up is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00064

Description	Name EthTrcvWakeupSleepOnDatalineEnabled Parent Container EthTrcvConfig Description Specifies if wake-up on data line according to OA TC10 is supported by the used Ethernet hardware (PHY). If the parameter is set to TRUE, wake-up on data line is supported according to OA TC10. Tags: atp.Status=draft Multiplicity 1 Type EcucBooleanParamDef Default value -- Post-Build Variant Value false Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local
Rejection Reason	Wake up is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00026

Description	Name EthTrcvIcuChannelRef Parent Container EthTrcvConfig Description Reference to the IcuChannel to enable/disable the interrupts for wakeups. Multiplicity 0..1 Type Symbolic name reference to [IcuChannel] Post-Build Variant Multiplicity false Post-Build Variant Value false Multiplicity Configuration Class
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	Pre-compile time X All Variants Link time -- Post-build time -- Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local
Rejection Reason	Wake up is not supported yet.

Rejected Requirement 1 SWS_EthTrcv_00040

Description	Container Name EthTrcvSwitchInterface Parent Container EthTrcvMgmtInterface Description This container includes the Switch interface configuration between an Ethernet Switch and an Ethernet Transceiver. If this container is configured the EthTrcv shall call EthSwt_WriteTrcvRegister / EthSwt_WriteTrcvRegister API to access the hardware ethernet transceiver. Post-Build Variant Multiplicity false Multiplicity Configuration Class Pre-compile time X All Variants Link time -- Post-build time --
Rejection Reason	Switch interface is not needed.

Rejected Requirement 1 SWS_EthTrcv_00042

Description	Name EthTrcvSwitchPortRef Parent Container EthTrcvSwitchInterface Description Reference to a switch port. Multiplicity 1 Type Symbolic name reference to [EthSwtPort] Post-Build Variant Value false Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local
Rejection Reason	Switch interface is not needed.

Rejected Requirement 1 SWS_EthTrcv_00041

Description	Name EthTrcvSwitchRef Parent Container EthTrcvSwitchInterface Description Reference to a switch configuration container. Multiplicity 1
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	Type Symbolic name reference to [EthSwtConfig] Post-Build Variant Value false Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local
Rejection Reason	Switch interface is not needed.

Rejected Requirement 1 SWS_EthTrcv_00054

Description	Name EthTrcvEnableCableDiagnosticApi Parent Container EthTrcvGeneral Description Enable/disable the APIs for cable diagnostic:EthTrcv_RunCableDiagnostic, EthTrcv_GetCableDiagnosticsResult Multiplicity 1 Type EcucBooleanParamDef Default value -- Post-Build Variant Value false Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local
Rejection Reason	Cable Diagnostic is not supported.

Rejected Requirement 1 SWS_EthTrcv_00055

Description	Name EthTrcvEnablePLCA Parent Container EthTrcvGeneral Description Enables the transmission with PLCA (Physical Layer Collision Avoidance) TRUE: PLCA enabled FALSE: PLCA disabled Tags: atp.Status=draft Multiplicity 1 Type EcucBooleanParamDef Default value false Post-Build Variant Value false Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local dependency: This parameter is relevant in case that EthTrcvPhysLayerType = TRCV_PHYS_LAYER_TYPE_10BASE_T1S. If PLCA mode is enabled via the parameter EthTrcvEnablePLCA the following parameter must
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	be configured: • EthTrcvPhysLayerPlcaLocalNodeID, • EthTrcvPhysLayerPlcaTransmitOpportunityTime, • EthTrcvPhysLayerPlcaNodeCount, • EthTrcvPhysLayerPlcaMaxBurstTime, • EthTrcvPhysLayerPlcaMaxBurstCount.
Rejection Reason	PLCA (Physical Layer Collision Avoidance) is not supported.

Rejected Requirement 1 SWS_EthTrcv_00061

Description	Name EthTrcvGetBusWuReasonApi Parent Container EthTrcvGeneral Description Enables / Disables API to obtain the recent wake-up reason detected by the used Ethernet hardware (e.g. PHY). Tags: atp.Status=draft Multiplicity 1 Type EcucBooleanParamDef Default value false Post-Build Variant Value false Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local
Rejection Reason	Wake up is not supported.

Rejected Requirement 1 SWS_EthTrcv_00031

Description	Name EthTrcvGetTransceiverWakeupModeApi Parent Container EthTrcvGeneral Description Enables / Disables EthTrcv_GetTransceiverWakeupMode API Multiplicity 0..1 Type EcucBooleanParamDef Default value -- Post-Build Variant Multiplicity false Post-Build Variant Value false Multiplicity Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Value Configuration Class Pre-compile time X All Variants Link time -- Post-build time -- Scope / Dependency scope: local dependency: Only valid if EthTrcvWakeUpSupport is not ETHTRCV_WAKEUP_NOT_SUPPORTED
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Rejection Reason	Wake up is not supported.
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Rejected Requirement 1 SWS_EthTrcv_00047

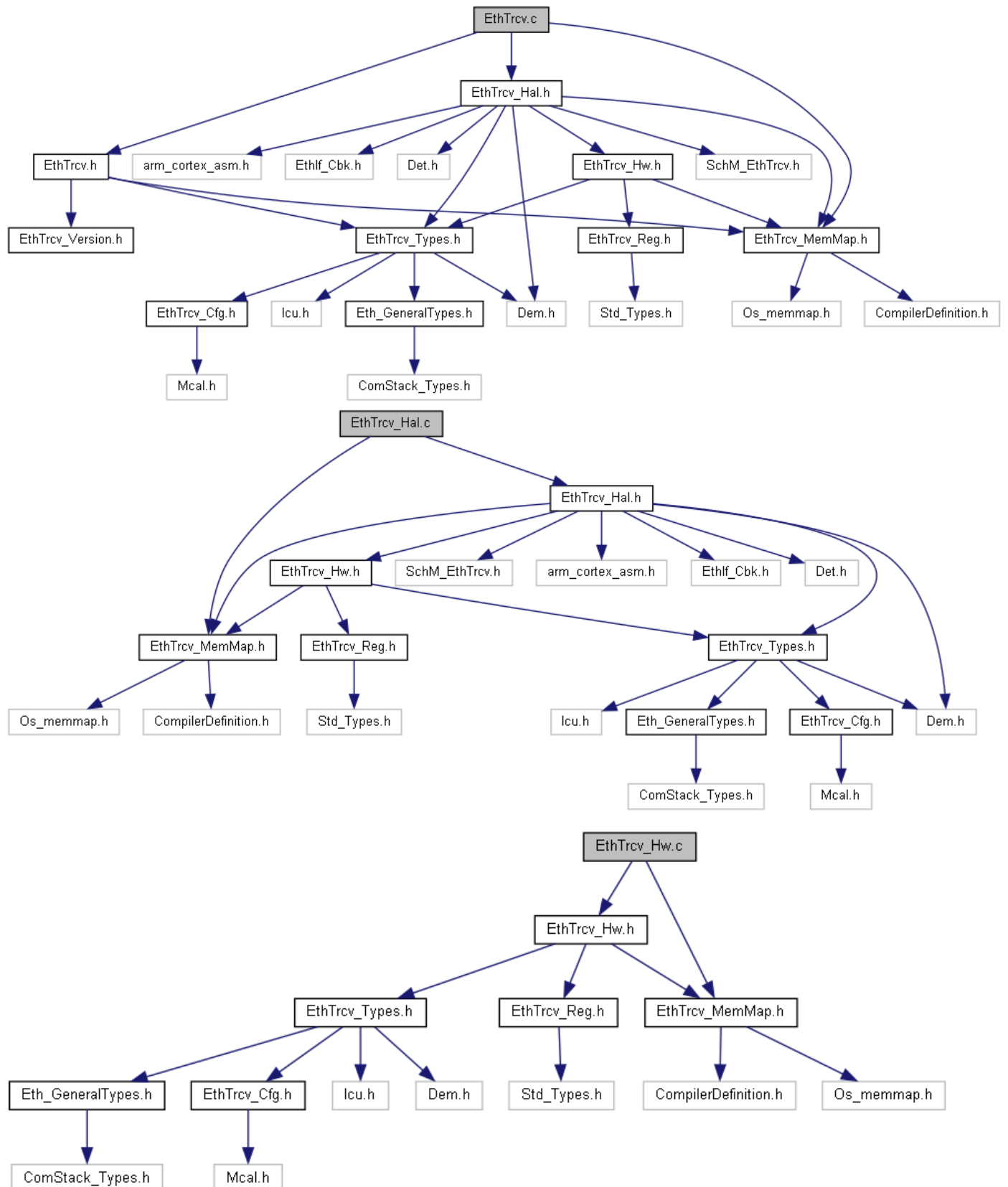
Description	Name EthTrcvSetPhyTestModeApi Parent Container EthTrcvGeneral Description Enables / Disables EthTrcv_SetPhyTestMode API. Multiplicity 1 Type EcucBooleanParamDef Default value -- Post-Build Variant Multiplicity false Post-Build Variant Value false Multiplicity Configuration Class Pre-compile time X AllVariants Link time -- Post-build time -- Value Configuration Class Pre-compile time X AllVariants Link time -- Post-build time -- Scope / Dependency scope: local
Rejection Reason	Phy Test Mode is not supported.

Rejected Requirement 1 SWS_EthTrcv_00030

Description	Name EthTrcvWakeUpSupport Parent Container EthTrcvGeneral Description Configures how to detect a signaled wake-up by hardware: polling, asynchronous check via signal, interrupt or to not used/not supported. In case detection of a wake-up is not support (e.g. wake-up detection not needed or Ethernet hardware does not support wake-up detection), the BSWMD pre-configuration shall be set to ETHTRCV_WAKEUP_NOT_SUPPORTED. Multiplicity 1 Type EcucEnumerationParamDef Range ETHTRCV_WAKEUP_BY_ASYNCHRONOUS_CHECK- Wake-up detection is done within the main function. Polling of the Ethernet hardware is only done, if a wake-up was signaled. Otherwise the Ethernet hardware is not checked for wake-up.Used e.g. if Ethernet hardware is OA TC10 compliant and the ECU maintain an Ethernet switch. ETHTRCV_WAKEUP_BY_INTERRUPT Wake-up detection is signaled by interrupt ETHTRCV_WAKEUP_BY_POLLING Wake-up detection is done by polling continuously within the main function ETHTRCV_WAKEUP_NOT_SUPPORTED Wake up is not supported Post-Build Variant Value false Value Configuration Class Pre-compile time X All Variants Link time --
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	Post-build time -- Scope / Dependency scope: local
Rejection Reason	Wake up is not supported.

2.2 File Structure



2.3 Macros

2.3.1 Macros in EthTrcv_Cfg.h

- **#define ETHTRCV_CFG_AR_RELEASE_MAJOR_VERSION 4**
- **#define ETHTRCV_CFG_AR_RELEASE_MINOR_VERSION 6**
- **#define ETHTRCV_CFG_AR_RELEASE_REVISION_VERSION 0**
- **#define ETHTRCV_CFG_SW_MAJOR_VERSION 1**
- **#define ETHTRCV_CFG_SW_MINOR_VERSION 1**
- **#define ETHTRCV_CFG_SW_PATCH_VERSION 0**
- **#define ETHTRCV_CFG_VENDOR_ID 174**
- **#define ETHTRCV_CFG_MODULE_ID 73**
- **#define ETHTRCV_PRECOMPILE_SUPPORT (STD_ON)**
- **#define ETHTRCV_DEV_ERROR_DETECT (STD_ON)**
EthTrcv Dev error detect switch.
- **#define ETHTRCV_ENABLE_CABLE_DIAGNOSTIC_API (STD_ON)**
Enable/disable the APIs for cable diagnostic.
- **#define ETHTRCV_ENABLE_PLCA (STD_OFF)**
Enables the transmission with PLCA (Physical Layer Collision Avoidance)
- **#define ETHTRCV_GET_BAUD_RATE_API (STD_ON)**
Enables / Disables EthTrcv_GetBaudRate API.
- **#define ETHTRCV_GET_BUS_WU_REASON_API (STD_ON)**
Enables / Disables API to obtain the recent wake-up reason detected by the used Ethernet hardware (e.g. PHY)
- **#define ETHTRCV_GET_DUPLEX_MODE_API (STD_ON)**
Enables / Disables EthTrcv_GetDuplexMode API
- **#define ETHTRCV_GET_LINK_STATE_API (STD_ON)**
Enables / Disables EthTrcv_GetLinkState API
- **#define ETHTRCV_GET_PHY_IDENTIFIER_API (STD_ON)**
Enables/Disables optional API EthTrcv_GetPhyIdentifier

- **#define ETHTRCV_GET_TRANSCEIVER_WAKEUP_MODE_API** (STD_OFF)
Enables / Disables EthTrcv_GetTransceiverWakeupMode API
- **#define ETHTRCV_INDEX** (0U)
Instanceld of this EthTrcvdriver
- **#define ETHTRCV_MAX_TRCVS_SUPPORTED** (1U)
Limits the total number of supported transceivers
- **#define ETHTRCV_SET_PHY_TEST_MODE_API** (STD_OFF)
Enables / Disables EthTrcv_SetPhyTestMode API
- **#define ETHTRCV_START_AUTONEGOTIATION_API** (STD_ON)
Enables / Disables EthTrcvStartAutoNegotiationApi API
- **#define ETHTRCV_VERSION_INFO_API** (STD_ON)
Enables / Disables version info API
- **#define ETHTRCV_VERSION_INFO_API_MACRO** (STD_OFF)
Enables / Disables version info API macro implementation
- **#define ETHTRCV_WAKEUP_NOT_SUPPORTED** (0U)
- **#define ETHTRCV_WAKEUP_BY_INTERRUPT** (1U)
- **#define ETHTRCV_WAKEUP_BY_POLLING** (2U)
- **#define ETHTRCV_WAKEUP_BY_ASYNCHRONOUS_CHECK** (3U)
- **#define ETHTRCV_WAKEUP_SUPPORT_TYPE** (ETHTRCV_WAKEUP_NOT_SUPPORTED)
Configures wake-up to polling or interrupt or to not used/not supported
- **#define ETHTRCV_DEM_EVENT_DETECT** (STD_ON)
Enables / Disables diagnostic Event
- **#define ETHTRCV_NUM_ECU_PARTITIONS** (3U)
The number of ECU partitions supported by EthTrcv driver
- **#define ETHTRCV_CHECK_ECU_PARTITION** (STD_ON)
Limits the min length bytes of FIFO buffers.

2.3.2 Macros in EthTrcv_Hw.h

- **#define ETHTRCV_INVALID_REG_VALUE** (0xFFFFU)
DMA TX descriptor Transmit Complete Interrupt Enable.
- **#define ETHTRCV_ORG_UNIQUE_ID (id1,id2)** (((uint32)(id1) & ETHTRCV_UTP_ID1_OUI_MASK) << ETHTRCV_UTP_ID1_OUI_SHIFT) | (((uint32)(id2) & ETHTRCV_UTP_ID2_OUI_MASK) >>

ETHTRCV_UTP_ID2_OUI_SHIFT))

Get the Organizationally Unique Identifier from transceiver ID1 and ID2 register

- #define **ETHTRCV_MODEL_NUMBER(id2)** (((id2) & ETHTRCV_UTP_ID2_MODEL_MASK) >> ETHTRCV_UTP_ID2_MODEL_SHIFT)
Get the Manufacturer's Model Number from transceiver ID2 register
- #define **ETHTRCV_REVISION_NUMBER(id2)** (((id2) & ETHTRCV_UTP_ID2_REVISION_MASK) >> ETHTRCV_UTP_ID2_REVISION_SHIFT)
Get the Revision Number from transceiver ID2 register

2.3.3 Macros in EthTrcv_Hal.h

- #define **EthTrcv_GetPartitionID()** GET_CPU_ID ()
- #define **ETHTRCV_CFG_PARTITION_ID(TrcvIdx,CoreId)** ((EthTrcv_Config_Ptrs[CoreId])->TrcvCfgPtr[TrcvIdx].EcuPartition)
- #define **ETHTRCV_CFG_NUM_TRANSCEIVERS(CoreId)** ((EthTrcv_Config_Ptrs[CoreId])->NumTransceivers)
- #define **ETHTRCV_CFG_TRCV_CFG_PTR(TrcvIdx,CoreId)** (&((EthTrcv_Config_Ptrs[CoreId])->TrcvCfgPtr[TrcvIdx]))
- #define **ETHTRCV_CFG_ETHIF_TRCVIDX(TrcvCfgPtr)** ((TrcvCfgPtr->EthIfTrcvIdx)
- #define **ETHTRCV_CFG_TRCV_AUTONEG_ENABLED(TrcvCfgPtr)** ((TrcvCfgPtr->ConnNeg == ETHTRCV_CONN_NEG_AUTO)
- #define **ETHTRCV_CFG_TRCV_SOFTRESET_ON_INIT(TrcvCfgPtr)** ((TrcvCfgPtr->SoftResetOnInit)
- #define **ETHTRCV_CFG_TRCV_SOFTRESET_READ_REPETITIONS (TrcvCfgPtr)** ((TrcvCfgPtr->SoftResetReadRepetitions)
- #define **ETHTRCV_CFG_TRCV_PHY_INTERFACE(TrcvCfgPtr)** ((TrcvCfgPtr->PhyIf)
- #define **ETHTRCV_CFG_TRCV_RGMII_TX_DELAY_ENABLED(TrcvCfgPtr)** ((TrcvCfgPtr->RgmiiTxDelayEnabled)
- #define **ETHTRCV_CFG_TRCV_RGMII_TX_DELAY_VALUE(TrcvCfgPtr)** ((TrcvCfgPtr->RgmiiTxDelayValue)
- #define **ETHTRCV_CFG_TRCV_RGMII_RX_DELAY_ENABLED(TrcvCfgPtr)** ((TrcvCfgPtr->RgmiiRxDelayEnabled)
- #define **ETHTRCV_CFG_TRCV_RGMII_RX_DELAY_VALUE(TrcvCfgPtr)** ((TrcvCfgPtr->RgmiiRxDelayValue)
- #define **ETHTRCV_CFG_TRCV_CLOCK_OUTPUT(TrcvCfgPtr)** ((TrcvCfgPtr->ClkOutPut)
- #define **ETHTRCV_CFG_TRCV_DUPLEX_MODE(TrcvCfgPtr)** ((TrcvCfgPtr->DuplexMode)
- #define **ETHTRCV_CFG_TRCV_SPEED_MODE(TrcvCfgPtr)** ((TrcvCfgPtr->MacLayerSpeed)

- `#define ETHTRCV_CFG_TRCV_AUTONEG_SPEED(TrcvCfgPtr) ((TrcvCfgPtr)->TrcvSpeed)`
- `#define ETHTRCV_CFG_TRCV_MII_INTERFACE(TrcvCfgPtr) ((TrcvCfgPtr)->MiifPtr)`
- `#define ETHTRCV_CFG_TRCV_DERIVATE(TrcvCfgPtr) ((TrcvCfgPtr)->DerivateType)`

2.3.4 Macros in EthTrcv_Version.h

- `#define ETHTRCV_VENDOR_ID 174`
- `#define ETHTRCV_MODULE_ID 73`
- `#define ETHTRCV_AR_RELEASE_MAJOR_VERSION 4`
- `#define ETHTRCV_AR_RELEASE_MINOR_VERSION 6`
- `#define ETHTRCV_AR_RELEASE_REVISION_VERSION 0`
- `#define ETHTRCV_SW_MAJOR_VERSION 1`
- `#define ETHTRCV_SW_MINOR_VERSION 1`
- `#define ETHTRCV_SW_PATCH_VERSION 0`

2.3.5 Macros in EthTrcv_Hal.c

- `#define ETHTRCV_TEST_MODE (3U)`
- `#define ETHTRCV_LOOPBACK_MODE (0x4U)`
MAC hash table index mask.
- `#define ETHTRCV_PHY_TX_MODE (5U)`
MAC hash table bit shift.

2.3.6 Macros in EthTrcv_.c

- `#define ETHTRCV_C_VENDOR_ID 174`
- `#define ETHTRCV_C_MODULE_ID 73`
- `#define ETHTRCV_C_AR_RELEASE_MAJOR_VERSION 4`
- `#define ETHTRCV_C_AR_RELEASE_MINOR_VERSION 6`
- `#define ETHTRCV_C_AR_RELEASE_REVISION_VERSION 0`
- `#define ETHTRCV_C_SW_MAJOR_VERSION 1`
- `#define ETHTRCV_C_SW_MINOR_VERSION 1`

- **#define ETHTRCV_C_SW_PATCH_VERSION 0**
- **#define ETHTRCV_DRV_SID_INIT 0x01**
- **#define ETHTRCV_DRV_SID_TRANSCEIVERLINKSTATEREQUEST 0x02**
- **#define ETHTRCV_DRV_SID_SETTRANSCEIVERMODE 0x03**
- **#define ETHTRCV_DRV_SID_GETTRANSCEIVERMODE 0x04**
- **#define ETHTRCV_DRV_SID_STARTAUTONEGOTIATION 0x05**
- **#define ETHTRCV_DRV_SID_GETLINKSTATE 0x06**
- **#define ETHTRCV_DRV_SID_GETBAUDRATE 0x07**
- **#define ETHTRCV_DRV_SID_GETDUPLEXMODE 0x08**
- **#define ETHTRCV_DRV_SID_READMIIINDICATION 0x09**
- **#define ETHTRCV_DRV_SID_WRITEMIIINDICATION 0x0A**
- **#define ETHTRCV_DRV_SID_GETVERSIONINFO 0x0B**
- **#define ETHTRCV_DRV_SID_MAINFUNCTION 0x0C**
- **#define ETHTRCV_DRV_SID_SETTRANSCEIVERWAKEUPMODE 0x0D**
- **#define ETHTRCV_DRV_SID_GETTRANSCEIVERWAKEUPMODE 0x0E**
- **#define ETHTRCV_DRV_SID_CHECKWAKEUP 0x0F**
- **#define ETHTRCV_DRV_SID_GETPHYSIGNALQUALITY 0x10**
- **#define ETHTRCV_DRV_SID_SETPHYTESTMODE 0x11**
- **#define ETHTRCV_DRV_SID_SETPHYLOOPBACKMODE 0x12**
- **#define ETHTRCV_DRV_SID_SETPHYTXMODE 0x13**
- **#define ETHTRCV_DRV_SID_GETCABLEDIAGNOSTICSRESULT 0x14**
- **#define ETHTRCV_DRV_SID_GETPHYIDENTIFIER 0x15**
- **#define ETHTRCV_DRV_SID_RUNCABLEDIAGNOSTIC 0x16**
- **#define ETHTRCV_DRV_SID_GETBUSWUREASON 0x17**

- #define **ETHTRCV_DRV_SID_GETMACMETHOD** 0x18
- #define **ETHTRCV_E_INV_CTRL_IDX** ((uint8)0x01)
- #define **ETHTRCV_E_UNINIT** ((uint8)0x02)
- #define **ETHTRCV_E_PARAM_POINTER** ((uint8)0x03)
- #define **ETHTRCV_E_NOT_SUPPORTED** ((uint8)0x04)
- #define **ETHTRCV_E_INV_PARTITION** ((uint8)0x05)

2.4 Enums

2.4.1 Enumerations in EthTrcv_Types.h

2.4.1.1 EthTrcv_ConnNegType

Enumeration	EthTrcv_ConnNegType	
Description	This type specify connection negotiation of the Transceiver.	
Values	Value	Description
	ETHTRCV_CONN_NEG_AUTO = 0x0U	Automatic Negotiation.
	ETHTRCV_CONN_NEG_MASTER = 0x1U	Master.
	ETHTRCV_CONN_NEG_NONE = 0x2U	PLCA.
	ETHTRCV_CONN_NEG_SLAVE = 0x3U	Slave.

2.4.1.2 EthTrcv_PhylfType

Enumeration	EthTrcv_PhylfType	
Description	This type specify PHY interface of the Transceiver.	
Values	Value	Description
	ETHTRCV_PHYIF_MII = 0U	MII interface.
	ETHTRCV_PHYIF_RMII = 1U	RMII interface.
	ETHTRCV_PHYIF_RGMII = 2U	RGMII interface.
	ETHTRCV_PHYIF_SGMII = 3U	SGMII interface.

2.4.1.3 EthTrcv_PhysLayerType

Enumeration	EthTrcv_PhysLayerType	
Description	This type specify physical layer of the Transceiver.	
Values	Value	Description
	ETHTRCV_PHYS_LAYER_TYPE_1000BASE_T = 0U	1000BASE_T
	ETHTRCV_PHYS_LAYER_TYPE_1000BASE_T1 = 1U	1000BASE_T1
	ETHTRCV_PHYS_LAYER_TYPE_100BASE_T1 = 2U	100BASE_T1
	ETHTRCV_PHYS_LAYER_TYPE_100BASE_TX = 3U	100BASE_Tx

	ETHTRCV_PHYS_LAYER_TYPE_10BASE_T1S = 4U	10BASE_T1S
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2.4.1.4 EthTrcv_DerivateType

Enumeration	EthTrcv_DerivateType	
Description	This type specify the derivate type of the Transceiver.	
Values	Value	Description
	ETHTRCV_YT8531SH= 0U	YT8531SH
	ETHTRCV_RTL8211FS = 1U	RTL8211FS
	ETHTRCV_RTL8211FS_RTL9010AA = 2U	RTL8211FS RTL9010AA bridge
	ETHTRCV_RTL8211FS_JL311X = 3U	RTL8211FS JL311X bridge
	ETHTRCV_YT8531SH_JL311X = 4U	YT8531SH JL311X bridge
	ETHTRCV_JL311X = 0x5U	JL311X
	ETHTRCV_JL311X_YT8531SH = 0x6U	JL311X YT8531SH bridge

2.4.1.5 EthTrcv_ClkOutPutType

Enumeration	EthTrcv_ClkOutPutType	
Description	This type specify physical layer of the Transceiver.	
Values	Value	Description
	ETHTRCV_CLK_OUTPUT_DISABLE = 0U	Clock output disabled
	ETHTRCV_CLK_OUTPUT_25M = 1U	Clock output 25M
	ETHTRCV_CLK_OUTPUT_125M = 2U	Clock output 125M

2.4.2 Enumerations in Eth_GeneralTypes.h

2.4.2.1 Eth_StateType

Enumeration	Eth_StateType	
Description	A variable of this type holds the state of the Ethernet driver module.	
Values	Value	Description
	ETH_STATE_UNINIT = 0U	The driver is not yet configured.
	ETH_STATE_INIT = 1U	The driver has been initialized.

2.4.2.2 Eth_ModeType

Enumeration	Eth_ModeType	
Description	This type indicates whether the Ethernet controller is stopped or running.	
Values	Value	Description
	ETH_MODE_DOWN = 0U	Disable the Ethernet Rx/Tx communication and set its corresponding hardware to a lowpower sleep mode and initiate a sleep process, if the Ethernet hardware provide such a feature. E.g. request a sleep on data line for OATC10 compatible Ethernet hardware.
	ETH_MODE_ACTIVE	Enable the Ethernet Rx/Tx communication and set its corresponding hardware to an power on mode.
	ETH_MODE_ACTIVE_WITH_WAKEUP_REQUEST	Enable the Ethernet Rx/Tx communication and set its corresponding hardware to an power on mode and

		request an wake-up on the network, if the Ethernet hardware provide a wake-up feature. E.g. wake-up on data line for OA TC10 compatible Ethernet hardware.
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2.4.2.3 Eth_RxStatusType

Enumeration	Eth_RxStatusType	
Description	The Ethernet reception status.	
Values	Value	Description
	ETH_RECEIVED = 0U	Ethernet frame has been received, no further frames available.
	ETH_NOT_RECEIVED	Ethernet frame has not been received, no further frames available.
	ETH_RECEIVED_MORE_DATA_AVAILABLE	Ethernet frame has been received, more frames are available.

2.4.2.4 Eth_FilterActionType

Enumeration	Eth_FilterActionType	
Description	Action type for MAC address filtering.	
Values	Value	Description
	ETH_ADD_TO_FILTER = 0U	Add MAC address to the filter.
	ETH_REMOVE_FROM_FILTER	Remove MAC address from the filter.

2.4.2.5 Eth_TimeStampQualType

Enumeration	Eth_TimeStampQualType	
Description	The Ethernet quality of timestamp type.	
Values	Value	Description
	ETH_VALID = 0U	Timestamp valid.
	ETH_INVALID	Timestamp invalid.
	ETH_UNCERTAIN	Timestamp uncertain.

2.4.2.6 EthTrcv_LinkStateType

Enumeration	EthTrcv_LinkStateType	
Description	This type defines the Ethernet link state.	
Values	Value	Description
	ETHTRCV_LINK_STATE_DOWN = 0U	No physical Ethernet connection established.
	ETHTRCV_LINK_STATE_ACTIVE	Physical Ethernet connection established.

2.4.2.7 EthTrcv_StateType

Enumeration	EthTrcv_StateType	
Description	This type defines the transceiver state.	
Values	Value	Description
	ETHTRCV_STATE_UNINIT = 0U	Driver is not yet configured.
	ETHTRCV_STATE_INIT	Driver is configured.

2.4.2.8 EthTrcv_BaudRateType

Enumeration	EthTrcv_BaudRateType	
Description	This type defines the Ethernet baud rate.	
Values	Value	Description
	ETHTRCV_BAUD_RATE_10MBIT = 0U	10MBIT Ethernet connection.
	ETHTRCV_BAUD_RATE_100MBIT	100MBIT Ethernet connection.
	ETHTRCV_BAUD_RATE_1000MBIT	1000MBIT Ethernet connection.
	ETHTRCV_BAUD_RATE_2500MBIT	2500MBIT Ethernet connection

2.4.2.9 EthTrcv_DuplexModeType

Enumeration	EthTrcv_DuplexModeType	
Description	This type defines the Ethernet duplex modes.	
Values	Value	Description
	ETHTRCV_DUPLEX_MODE_HALF = 0U	Half duplex Ethernet connection.
	ETHTRCV_DUPLEX_MODE_FULL	Full duplex Ethernet connection.

2.4.2.10 EthTrcv_WakeupModeType

Enumeration	EthTrcv_WakeupModeType	
Description	This type controls the transceiver wake up modes and/or clears the wake-up reason.	
Values	Value	Description
	ETHTRCV_WUM_DISABLE = 0U	Transceiver wake up disabled.
	ETHTRCV_WUM_ENABLE	Transceiver wake up enabled.
	ETHTRCV_WUM_CLEAR	Transceiver wake up reason cleared.

2.4.2.11 EthTrcv_WakeupReasonType

Enumeration	EthTrcv_WakeupReasonType	
Description	This type defines the transceiver wake up reasons.	
Values	Value	Description
	ETHTRCV_WUR_NONE = 0U	No wake up reason detected.
	ETHTRCV_WUR_GENERAL	General wake up detected, no distinct reason supported by hardware.
	ETHTRCV_WUR_BUS	Bus wake up detected. Available if supported by hardware.
	ETHTRCV_WUR_INTERNAL	Internal wake up detected. Available if supported by hardware.
	ETHTRCV_WUR_RESET	Reset wake up detected. Available if supported by hardware.
	ETHTRCV_WUR_POWER_ON	Power on wake up detected. Available if supported by hardware.
	ETHTRCV_WUR_PIN	Pin wake up detected. Available if supported by hardware.
	ETHTRCV_WUR_SYSERR	System error wake up detected. Available if supported by hardware.
	ETHTRCV_WUR_WODL_WUP	Wake-up on data line (WUP = wake up pulse) detected while link is down of the corresponding Ethernet hardware e.g. PHY). Only available if EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE.

	ETHTRCV_WUR_WODL_WUR	Wake-up on data line (WUR = wake up request)) detected while link is active of the used Ethernet hardware (e.g. PHY). Only available if EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE.
	ETHTRCV_WUR_TRANSFER	Device internal transfer of wake-up on data line from a neighboring PHY. Only possible if Ethernet hardware is compliant to OA TC10.

2.4.2.12 EthTrcv_PhyTestModeType

Enumeration	EthTrcv_PhyTestModeType	
Description	Describes the possible PHY test modes.	
Values	Value	Description
	ETHTRCV_PHYTESTMODE_NONE = 0U	Normal operation.
	ETHTRCV_PHYTESTMODE_1	Test transmitter droop.
	ETHTRCV_PHYTESTMODE_2	Test master timing jitter.
	ETHTRCV_PHYTESTMODE_3	Test slave timing jitter.
	ETHTRCV_PHYTESTMODE_4	Test transmitter distortion.
	ETHTRCV_PHYTESTMODE_5	Test power spectral density (PSD) mask.

2.4.2.13 EthTrcv_PhyLoopbackModeType

Enumeration	EthTrcv_PhyLoopbackModeType	
Description	Describes the possible PHY loopback modes.	
Values	Value	Description
	ETHTRCV_PHYLOOPBACK_NONE = 0U	Normal operation.
	ETHTRCV_PHYLOOPBACK_INTERNAL	Internal loopback.
	ETHTRCV_PHYLOOPBACK_EXTERNAL	External loopback.
	ETHTRCV_PHYLOOPBACK_REMOTE	Remote loopback.

2.4.2.14 EthTrcv_PhyTxModeType

Enumeration	EthTrcv_PhyTxModeType	
Description	Describes the possible PHY transmit modes.	
Values	Value	Description
	ETHTRCV_PHYTXMODE_NORMAL = 0U	Normal operation.
	ETHTRCV_PHYTXMODE_TX_OFF	Transmitter disabled.
	ETHTRCV_PHYTXMODE_SCRAMBLER_OFF	Scrambler disabled.

2.4.2.15 EthTrcv_CableDiagResultType

Enumeration	EthTrcv_CableDiagResultType	
Description	Describes the results of the cable diagnostics.	
Values	Value	Description
	ETHTRCV_CABLEDIAG_OK = 0U	Cable diagnostic ok.
	ETHTRCV_CABLEDIAG_ERROR	Cable diagnostic failed.
	ETHTRCV_CABLEDIAG_SHORT	Short circuit detected.
	ETHTRCV_CABLEDIAG_OPEN	Open circuit detected.

	ETHTRCV_CABLEDIAG_PENDING	Cable diagnostic is still running.
	ETHTRCV_CABLEDIAG_WRONG_POLARITY	Cable diagnostics has detected wrong polarity of the "Ethernet physical+" or "Ethernet physical-" lines.

2.4.2.16 EthTrcv_MacMethodType

Enumeration	EthTrcv_MacMethodType	
Description	This type defines the media access control type in half duplex.	
Values	Value	Description
	ETHTRCV_MAC_TYPE_CSMA_CD = 0U	Carrier sense multiple access with collision detection.
	ETHTRCV_MAC_TYPE_PLCA	Physical layer collision avoidance.

2.4.2.17 EthSwt_StateType

Enumeration	EthSwt_StateType	
Description	Status supervision used for Development Error Detection.	
Values	Value	Description
	ETHSWT_STATE_UNINIT = 0U	The Switch is not yet configured.
	ETHSWT_STATE_INIT	The Switch is configured.
	ETHSWT_STATE_PORTINIT_COMPLETED	Port initialization is completed.
	ETHSWT_STATE_ACTIVE	The Switch is active.

2.4.2.18 EthSwt_MacLearningType

Enumeration	EthSwt_MacLearningType	
Description	This type defines the MAC addresses learning modes.	
Values	Value	Description
	ETHSWT_MACLEARNING_HWDISABLED = 0U	If hardware learning disabled, the switch must not learn new MAC addresses.
	ETHSWT_MACLEARNING_HWENABLED	If hardware learning enabled, the switch learns new MAC addresses.
	ETHSWT_MACLEARNING_SWENABLED	If software learning enabled, the hardware learning is disabled and the switch forwards packets with an unknown source address to a host CPU.

2.4.2.19 EthSwt_PortMirrorStateType

Enumeration	EthSwt_PortMirrorStateType	
Description	It shall be possible to set and get the mirror state (enable / disable) of the ethernet switch port.	
Values	Value	Description
	ETHSWT_PORT_MIRROR_DISABLED = 0U	Port mirroring disabled.
	ETHSWT_PORT_MIRROR_ENABLED	Port mirroring enabled.
	ETHSWT_PORT_MIRRORING_CONFIGURATION_NOT_SUPPORTED	Port mirroring configuration is not supported by Ethernet switch driver or by the Ethernet switch hardware.

2.4.2.20 EthSwt_MgmtOwner

Enumeration	EthSwt_MgmtOwner	
Description	Holds information if upper layer or EthSwt is owner of mgmt_obj.	

Values	Value	Description
	ETHSWT_MGMT_OBJ_UNUSED = 0U	Object unused.
	ETHSWT_MGMT_OBJ_OWNED_BY_ETHSWT	Object used and EthSwt collects needed data.
	ETHSWT_MGMT_OBJ_OWNED_BY_UPPER_LAYER	Object used and the upper layer does calculations.

2.5 Typedefs

2.5.1 Typedefs in Eth_GeneralTypes.h

- typedef uint16 **Eth_FrameType**
This type is used to pass the uint16 value of type or length field in the Ethernet frame header.
- typedef uint8 **Eth_DataType**
Type used to pass transmit or receive data to or from the driver.
- typedef uint32 **Eth_BufIdxType**
Indicate the index of buffer used in transmitted and received data.

2.5.2 Typedefs in EthTrcv_Types.h

- typedef void (*EthTrcv_WakeUpCalloutType)(uint8 TrcvIdx)
This type is used to define the wake up call function.

2.6 Structures

2.6.1 EthTrcv_ConfigType

Structure	EthTrcv_ConfigType
Description	Implementation specific structure of the Ethernet Transceiver configuration.
Diagram	<pre> graph BT A[EthTrcv_TrcvCfgType] --> B[EthTrcv_ConfigType] B --> C[EthTrcv_ConfigType_struct] </pre>
Data Fields	• uint8 NumTransceivers The number of enabled transceivers. • const EthTrcv_TrcvCfgType * TrcvCfgPtr Transceiver configurations.

2.6.2 EthTrcv_TrcvCfgType

Structure	EthTrcv_TrcvCfgType
Description	Implementation specific structure of the Ethernet Transceiver configuration.

Diagram	<pre> graph TD EthTrcv_SwtIfType -- SwtIfPtr --> EthTrcv_TrcvCfgType EthTrcv_MiIfType -- MiIfPtr --> EthTrcv_TrcvCfgType EthTrcv_TrcvCfgType -- TrcvCfgPtr --> EthTrcv_ConfigType EthTrcv_ConfigType --> EthTrcv_ConfigType_struct </pre>
Data Fields	• Uint8 EcuPartition The referenced ECUC partition ID. • uint8 EthIfTrcvIdx Index of the transceiver within the context of the Ethernet Interface. • boolean ActAsSlavePassiveEnabled Acting as a passive communication slave on the corresponding ComM channel. • boolean ForceSleepEnabled Specifies if the Ethernet hardware (PHY) support to go to sleep without the confirmation of the Ethernet hardware (PHY) of the linked Ethernet communication partner. • boolean SoftResetOnInit Specifies if software reset is enabled (TRUE) or disabled (FALSE) on initialization. • uint16 SoftResetReadRepetitions The repetitions to wait after a software reset. • EthTrcv_ConnNegType ConnNeg The connection negotiation of the Ethernet transceiver link. • EthTrcv_DuplexModeType DuplexMode The duplex mode of the Ethernet transceiver link. • EthTrcv_BaudRateType MacLayerSpeed Defines the baud rate of the MAC layer. • EthTrcv_PhylfType Phylf PHY interface connected to ECU. • EthTrcv_PhysLayerType PhysLayer Specifies the physical layer type of the transceiver link. • uint8 PlcaLocalNodeId Configuration parameter for the transceiver node ID when the PLCA mode for 10BASE-T1S is used. • uint8 PlcaMaxBurstCount Defines maximum packets allowed to be transmitted within a TO. The maximum burst count could differ per ECU within a PLCA mixed segment. • uint8 PlcaMaxBurstTimer Limits the burst frames in bit time. The maximum burst time could differ per ECU within a PLCA mixed segment. • PlcaNodeCount Defines the number of communication participants on the mixed segment. • uint8 PlcaTransmitOpportunityTimer Timer for the transmission in bit time to evaluate if a Transmission Opportunity is yield or not. • uint8 SleepRequestNumberOfRepetitions Specifies the repetitions to trigger a Sleep.

- uint16 SleepRequestRepetitionPeriod
Specifies the repetition period in seconds of repetitions for a Sleep Request.
- uint16 SleepModeExecutionDelay
Specifies the time delay in seconds to execute a sleep (see OA TC10) for a Ethernet hardware (PHY).
- EthTrcv_BaudRateType TrcvSpeed
Specifies the speed of the Ethernet transceiver link in [MBit/s]. If AutoNegotiation is enabled.
- boolean WakeupLocalEnabled
Specifies if local wake-up is enabled or disabled.
- boolean WakeupRemoteEnabled
Specifies if remote wake-up is enabled or disabled.
- boolean WakeupForwardLocalEnabled
Specifies if remote wake up forwarding is enabled (TRUE) or disabled (FALSE) .
- boolean WakeupForwardRemoteEnabled
Specifies if local wake up forwarding is enabled (TRUE) or disabled (FALSE) .
- uint32 WakeupLocalDetectionTime
Defines the time in seconds when a local wake-up.
- uint32 WakeupLocalDurationTime
Defines the duration time in seconds how long a local wake-up should be present on the local wake-up connection.
- boolean WakeupSleepOnDatalineEnabled
Specifies if wake-up on data line according to OA TC10 is supported by the used Ethernet hardware (PHY).
- uint8 WakeupRequestNumberOfRepetitions
Specifies the repetitions to trigger a wake-up request.
- uint16 WakeUpRequestRepetitionPeriod
Specifies the repetition period in seconds of a wake-up request.
- EthTrcv_DerivateType DerivateType
Specifies the transceiver chip derivate type.
- boolean RgmiiTxDelayEnabled
Enable or disable the TX clock delay in RGMII mode.
- uint8 RgmiiTxDelayValue
Specifies the TX clock delay value in RGMII mode.
- boolean RgmiiRxDelayEnabled
Enable or disable the RX clock delay in RGMII mode.
- uint8 RgmiiRxDelayValue
Specifies the RX clock delay value in RGMII mode.
- EthTrcv_ClkOutPutType ClkOutPut
Clock output mode of the transceiver.
- const EthTrcv_MiilfType *MiilfPtr
The MII interface configuration between an Ethernet Controller and the Ethernet Transceiver.
- const EthTrcv_SwtIfType *SwtIfPrt
The switch interface configuration between an Ethernet Controller and the Ethernet Transceiver.
- EthTrcv_WakeUpCalloutType WakeupCallout
Configuration of the call-out name.
- Icu_ChannelType IcuChannel
Reference to the IcuChannel to enable/disable the interrupts for wakeups.
- Mcal_DemErrorType Cfg_Dem_EAccess

DEM access error.

2.6.3 EthTrcv_MiilfType

Structure	EthTrcv_MiilfType
Description	This type specify the Mii interface information including MII(clause 22) and MMD(clause 45) callback functions Implemented by Eth driver, Eth controller index, transceiver index,extra bridged transceiver index and MII Pin multiplexing function.
Diagram	N/A
Data Fields	- Std_ReturnType (*WriteMii)(uint8 CtrlIdx,uint8 TrcvIdx,uint8 RegIdx,uint16 RegVal) - Std_ReturnType (*ReadMii)(uint8 CtrlIdx,uint8 TrcvIdx,uint8 RegIdx,uint16* RegValPtr) - Std_ReturnType (*WriteMmd)(uint8 CtrlIdx,uint8 TrcvIdx,uint8 Mmd,uint16 RegIdx,uint16 RegVal) - Std_ReturnType (*ReadMmd)(uint8 CtrlIdx,uint8 TrcvIdx,uint8 Mmd,uint16 RegIdx,uint16* RegValPtr) - uint8 CtrlIdx The Eth controller index used for MII access to the transceiver. - uint8 Miildx The transceiver index used for MII access to the transceiver. - uint8 ExtraMiildx The transceiver index used for MII access to the extra bridged transceiver. - void (*MiiMux)(boolean ExtraPhy) The Mii pins multiplexing call back function.

2.6.4 EthTrcv_SwtIfType

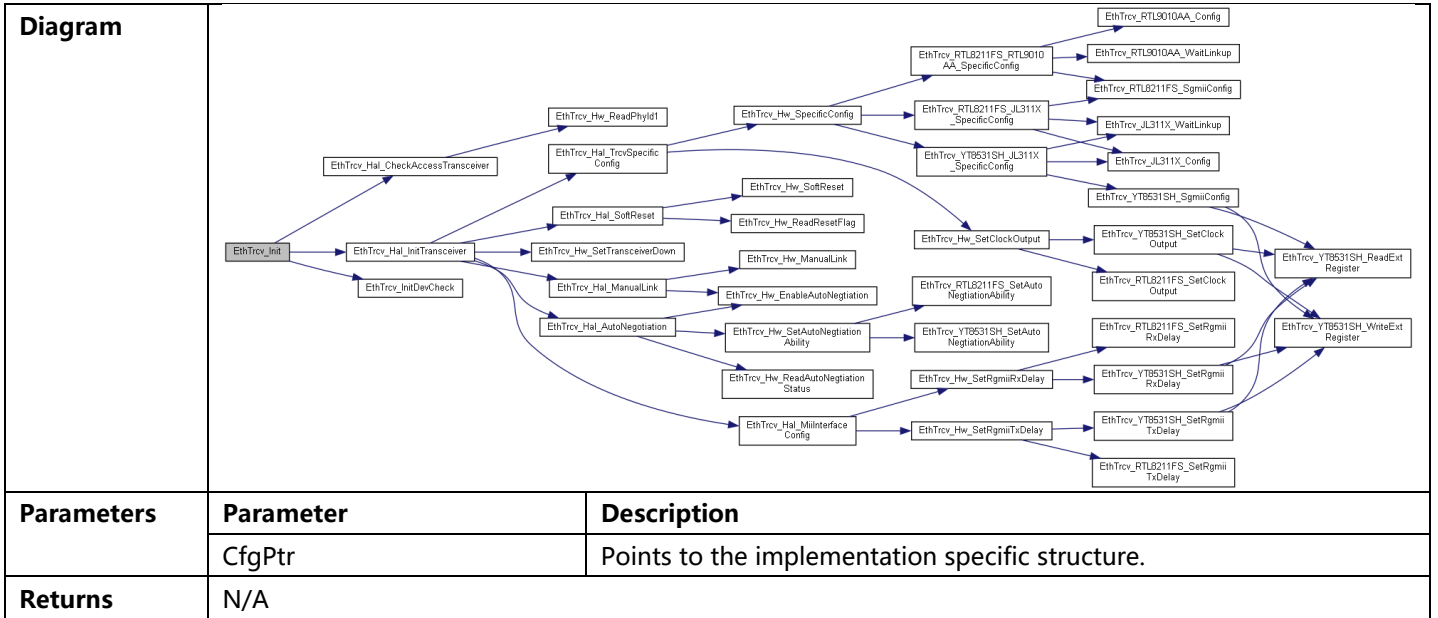
Structure	EthTrcv_SwtIfType
Description	The switch interface information.
Diagram	N/A
Data Fields	- Std_ReturnType (*SwtWrite)(uint8 SwitchIdx,uint8 SwitchPortIdx,uint8 RegIdx,uint16 RegVal) - Std_ReturnType (*SwtRead)(uint8 SwitchIdx,uint8 SwitchPortIdx,uint8 RegIdx,uint16* RegValPtr) - uint8 SwitchIdx Reference to a switch configuration container. - uint8 SwitchPortIdx Reference to a switch port.

2.7 API Functions

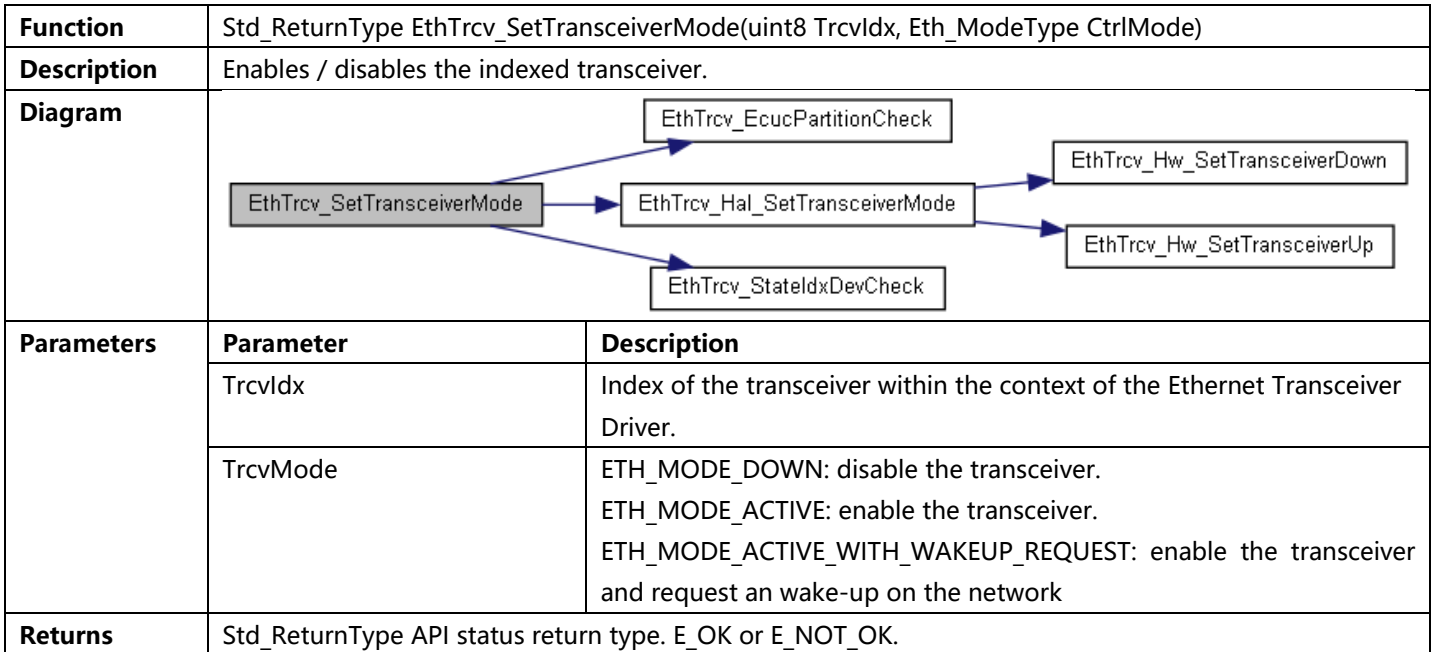
2.7.1 Functions in EthTrcv.h

2.7.1.1 void EthTrcv_Init(const EthTrcv_ConfigType* CfgPtr)

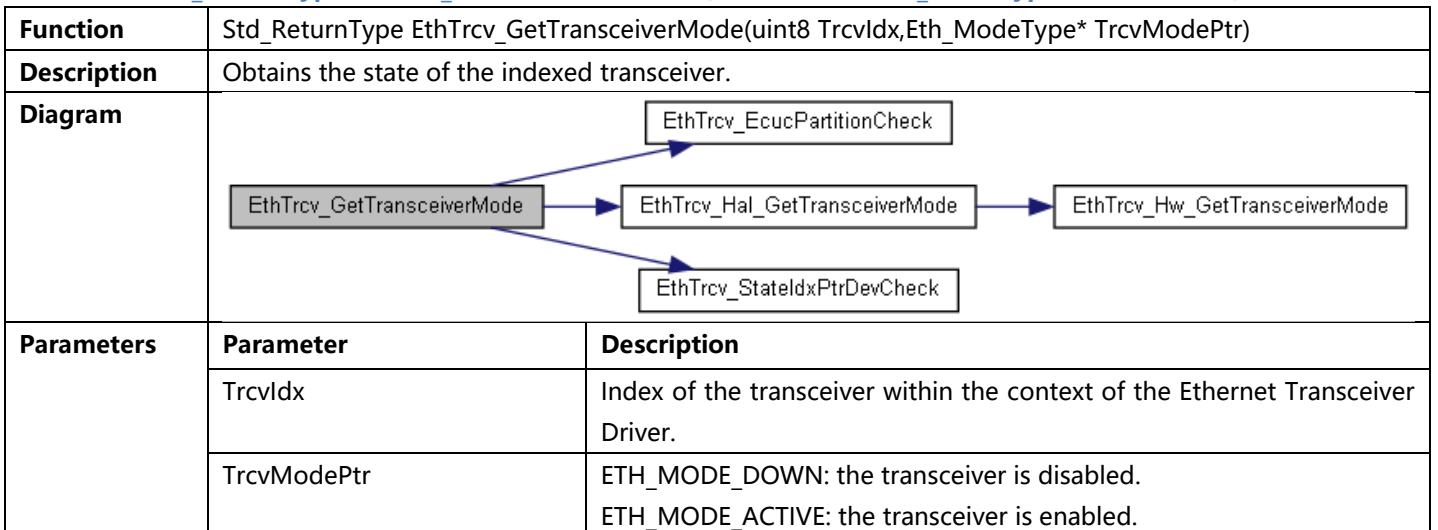
Function	void EthTrcv_Init(const EthTrcv_ConfigType* CfgPtr)
Description	Initializes the Ethernet Transceiver Driver



2.7.1.2 Std_ReturnType EthTrcv_SetTransceiverMode(uint8 TrcvIdx, Eth_ModeType CtrlMode)



2.7.1.3 Std_ReturnType EthTrcv_GetTransceiverMode(uint8 TrcvIdx, Eth_ModeType* TrcvModePtr)



Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.
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2.7.1.4 Std_ReturnType EthTrcv_SetTransceiverWakeupMode(uint8 TrcvIdx, EthTrcv_WakeupModeType TrcvWakeupMode)

Function	Std_ReturnType EthTrcv_SetTransceiverWakeupMode(uint8 TrcvIdx, EthTrcv_WakeupModeType TrcvWakeupMode)	
Description	Enables / disables the wake-up mode or clear the wake-up reason of the indexed transceiver.	
Diagram	<pre> graph LR A[EthTrcv_SetTransceiverWakeupMode] --> B[EthTrcv_EcucPartitionCheck] A --> C[EthTrcv_Hal_SetTransceiverWakeupMode] A --> D[EthTrcv_StateIdxDevCheck] </pre>	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.
	TrcvWakeupMode	ETHTRCV_WUM_DISABLE: disable transceiver wake up. ETHTRCV_WUM_ENABLE: enable transceiver wake up. ETHTRCV_WUM_CLEAR: clears transceiver wake up reason.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.5 Std_ReturnType EthTrcv_GetTransceiverWakeupMode(uint8 TrcvIdx, EthTrcv_WakeupModeType* TrcvWakeupModePtr)

Function	Std_ReturnType EthTrcv_GetTransceiverWakeupMode(uint8 TrcvIdx, EthTrcv_WakeupModeType* TrcvWakeupModePtr)	
Description	Obtains the wake up mode of the indexed transceiver.	
Diagram	<pre> graph LR A[EthTrcv_GetTransceiverWakeupMode] --> B[EthTrcv_EcucPartitionCheck] A --> C[EthTrcv_Hal_GetTransceiverWakeupMode] A --> D[EthTrcv_StateIdxPtrDevCheck] </pre>	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.
	TrcvWakeupModePtr	ETHTRCV_WUM_DISABLE: transceiver wake up is disabled. ETHTRCV_WUM_ENABLE: transceiver wake up is enabled.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.6 Std_ReturnType EthTrcv_GetBusWuReason(uint8 TrcvIdx, EthTrcv_WakeupReasonType* WakeupReasonPtr)

Function	Std_ReturnType EthTrcv_GetBusWuReason(uint8 TrcvIdx, EthTrcv_WakeupReasonType* WakeupReasonPtr)
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Description	This function returns the least recent wakeup reasons.	
Diagram	<pre> graph LR A[EthTrcv_GetBusWuReason] --> B[EthTrcv_EcucPartitionCheck] A --> C[EthTrcv_Hal_GetBusWuReason] A --> D[EthTrcv_StateIdxPtrDevCheck] </pre>	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.
	WakeupReasonPtr	Pointer to structure of least recent wakeup event, which was detected by the Ethernet PHY.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.7 Std_ReturnType EthTrcv_CheckWakeup(uint8 TrcvIdx)

Function	Std_ReturnType EthTrcv_CheckWakeup(uint8 TrcvIdx)	
Description	Service is called by EthIf in case a wake-up interrupt is detected.	
Diagram	<pre> graph LR A[EthTrcv_CheckWakeup] --> B[EthTrcv_EcucPartitionCheck] A --> C[EthTrcv_Hal_CheckWakeup] A --> D[EthTrcv_StateIdxDevCheck] </pre>	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.8 Std_ReturnType EthTrcv_StartAutoNegotiation (uint8 TrcvIdx)

Function	Std_ReturnType EthTrcv_StartAutoNegotiation (uint8 TrcvIdx)	
Description	Restarts the negotiation of the transmission parameters used by the indexed transceiver.	
Diagram	<pre> graph LR A[EthTrcv_StartAutoNegotiation] --> B[EthTrcv_EcucPartitionCheck] A --> C[EthTrcv_Hal_StartAutoNegotiation] A --> D[EthTrcv_StateIdxDevCheck] C --> E[EthTrcv_Hw_RestartAutoNegtiation] </pre>	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.9 Std_ReturnType EthTrcv_TransceiverLinkStateRequest(uint8 TrcvIdx,EthTrcv_LinkStateType LinkState)

Function	Std_ReturnType EthTrcv_TransceiverLinkStateRequest(uint8 TrcvIdx,EthTrcv_LinkStateType LinkState)	
Description	Request the given link state for the given Ethernet transceiver.	

Diagram	<pre>graph LR; A[EthTrcv_TransceiverLinkStateRequest] --> B[EthTrcv_EcucPartitionCheck]; A --> C[EthTrcv_Hal_TransceiverLinkStateRequest]; A --> D[EthTrcv_StateIdxDevCheck];</pre>							
Parameters	<table><tr><th>Parameter</th><th>Description</th></tr><tr><td>TrcvIdx</td><td>Index of the transceiver within the context of the Ethernet Transceiver Driver.</td></tr><tr><td>LinkState</td><td>The Ethernet link state of a physical Ethernet connection.</td></tr></table>	Parameter	Description	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.	LinkState	The Ethernet link state of a physical Ethernet connection.	
Parameter	Description							
TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.							
LinkState	The Ethernet link state of a physical Ethernet connection.							
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.							

2.7.1.10 Std_ReturnType EthTrcv_GetLinkState (uint8 TrcvIdx, EthTrcv_LinkStateType LinkStatePtr)

Function	Std_ReturnType EthTrcv_GetLinkState (uint8 TrcvIdx,EthTrcv_LinkStateType LinkStatePtr)	
Description	Obtains the link state of the indexed transceiver.	
Diagram	<pre>graph LR; A[EthTrcv_GetLinkState] --> B[EthTrcv_EcucPartitionCheck]; A --> C[EthTrcv_Hal_GetLinkState]; A --> D[EthTrcv_StateIdxPtrDevCheck]; C --> E[EthTrcv_Hw_ReadLinkState]; E --> F[EthTrcv_JL311X_ReadLinkState]; E --> G[EthTrcv_RTL8211FS_ReadLinkState]; E --> H[EthTrcv_RTL9010AA_ReadLinkState]; E --> I[EthTrcv_YT8531SH_ReadLinkState];</pre>	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.
	LinkStatePtr	ETHTRCV_LINK_STATE_DOWN: transceiver is disconnected. ETHTRCV_LINK_STATE_ACTIVE: transceiver is connected.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.11 Std_ReturnType EthTrcv_GetBaudRate (uint8 TrcvIdx, EthTrcv_BaudRateType* BaudRatePtr)

Function	Std_ReturnType EthTrcv_GetBaudRate (uint8 TrcvIdx, EthTrcv_BaudRateType* BaudRatePtr)	
Description	Obtains the baud rate of the indexed transceiver.	
Diagram	<pre>graph LR; A[EthTrcv_GetBaudRate] --> B[EthTrcv_EcucPartitionCheck]; A --> C[EthTrcv_Hal_GetBaudRate]; A --> D[EthTrcv_StateIdxPtrDevCheck]; C --> E[EthTrcv_Hw_ReadBaudRate]; E --> F[EthTrcv_JL311X_ReadBaudRate]; E --> G[EthTrcv_RTL8211FS_ReadBaudRate]; E --> H[EthTrcv_RTL9010AA_ReadBaudRate]; E --> I[EthTrcv_YT8531SH_ReadBaudRate];</pre>	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.
	BaudRatePtr	ETHTRCV_BAUD_RATE_10MBIT: 10MBit connection. ETHTRCV_BAUD_RATE_100MBIT: 100MBit connection. ETHTRCV_BAUD_RATE_1000MBIT: 1000MBit connection.

	ETHTRCV_BAUD_RATE_2500MBIT: 2500MBIT Ethernet connection.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.

2.7.1.12 Std_ReturnType EthTrcv_GetDuplexMode (uint8 TrcvIdx, EthTrcv_DuplexModeType* DuplexModePtr)

Function	Std_ReturnType EthTrcv_GetDuplexMode (uint8 TrcvIdx, EthTrcv_DuplexModeType* DuplexModePtr)	
Description	Obtains the link state of the indexed transceiver.	
Diagram	<pre> graph LR EthTrcv_GetDuplexMode --> EthTrcv_EcucPartitionCheck EthTrcv_GetDuplexMode --> EthTrcv_Hal_GetDuplexMode EthTrcv_GetDuplexMode --> EthTrcv_StateIdxPtrDevCheck EthTrcv_Hal_GetDuplexMode --> EthTrcv_Hw_ReadDuplexMode EthTrcv_Hw_ReadDuplexMode --> EthTrcv_JL311X_ReadDuplexMode EthTrcv_Hw_ReadDuplexMode --> EthTrcv_RTL8211FS_ReadDuplexMode EthTrcv_Hw_ReadDuplexMode --> EthTrcv_RTL9010AA_ReadDuplexMode EthTrcv_Hw_ReadDuplexMode --> EthTrcv_YT8531SH_ReadDuplexMode </pre>	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.
	DuplexModePtr	ETHTRCV_DUPLEX_MODE_HALF: half duplex connections. ETHTRCV_DUPLEX_MODE_FULL: full duplex connection.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.13 Std_ReturnType EthTrcv_SetPhyTestMode(uint8 TrcvIdx, EthTrcv_PhyTestModeType Mode)

Function	Std_ReturnType EthTrcv_SetPhyTestMode(uint8 TrcvIdx, EthTrcv_PhyTestModeType Mode)	
Description	Activates a given test mode.	
Diagram	<pre> graph LR EthTrcv_SetPhyTestMode --> EthTrcv_EcucPartitionCheck EthTrcv_SetPhyTestMode --> EthTrcv_Hal_SetPhyTestMode EthTrcv_SetPhyTestMode --> EthTrcv_StateIdxDevCheck EthTrcv_Hal_SetPhyTestMode --> EthTrcv_Hw_SetTestMode EthTrcv_Hal_SetPhyTestMode --> EthTrcv_Hw_TestModeSupported </pre>	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.
	Mode	Test mode to be activated.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.14 Std_ReturnType EthTrcv_SetPhyLoopbackMode(uint8 TrcvIdx, EthTrcv_PhyLoopbackModeType Mode)

Function	Std_ReturnType EthTrcv_SetPhyLoopbackMode(uint8 TrcvIdx, EthTrcv_PhyLoopbackModeType Mode)	
Description	Activates a given loopback mode.	
Diagram	<pre> graph LR EthTrcv_SetPhyLoopbackMode --> EthTrcv_EcucPartitionCheck EthTrcv_SetPhyLoopbackMode --> EthTrcv_Hal_SetPhyLoopbackMode EthTrcv_SetPhyLoopbackMode --> EthTrcv_StateIdxDevCheck EthTrcv_Hal_SetPhyLoopbackMode --> EthTrcv_Hw_SetLoopbackMode </pre>	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.

	Mode	Loopbackmode to be activated.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.15 Std_ReturnType EthTrcv_GetPhySignalQuality(uint8 TrcvIdx, uint32* SignalQualityPtr)

Function	Std_ReturnType EthTrcv_GetPhySignalQuality(uint8 TrcvIdx, uint32* SignalQualityPtr)	
Description	Obtains the current signal quality of the link of the indexed transceiver.	
Diagram	<pre> graph LR A[EthTrcv_GetPhySignalQuality] --> B[EthTrcv_EcucPartitionCheck] A --> C[EthTrcv_Hal_GetPhySignalQuality] A --> D[EthTrcv_StateIdxPtrDevCheck] C --> E[EthTrcv_Hw_GetSignalQuality] C --> F[EthTrcv_Hw_SignalQualitySupported] </pre>	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.
	SignalQualityPtr	Pointer to the memory where the signal quality shall be stored.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.16 Std_ReturnType EthTrcv_SetPhyTxMode(uint8 TrcvIdx, EthTrcv_PhyTxModeType Mode)

Function	Std_ReturnType EthTrcv_SetPhyTxMode(uint8 TrcvIdx, EthTrcv_PhyTxModeType Mode)	
Description	Activates a given transmission mode.	
Diagram	N/A	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.
	Mode	Transmission mode to be activated.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.17 Std_ReturnType EthTrcv_RunCableDiagnostic (uint8 TrcvIdx)

Function	Std_ReturnType EthTrcv_RunCableDiagnostic (uint8 TrcvIdx)	
Description	Trigger the cable diagnostics for the given Ethernet transceiver.	
Diagram	<pre> graph LR A[EthTrcv_RunCableDiagnostic] --> B[EthTrcv_EcucPartitionCheck] A --> C[EthTrcv_Hal_RunCableDiagnostic] A --> D[EthTrcv_StateIdxDevCheck] </pre>	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.18 Std_ReturnType EthTrcv_GetCableDiagnosticsResult(uint8 TrcvIdx, EthTrcv_CableDiagResultType* ResultPtr)

Function	Std_ReturnType EthTrcv_GetCableDiagnosticsResult(uint8 TrcvIdx, EthTrcv_CableDiagResultType* ResultPtr)	
Description	Retrieves the cable diagnostics result of a given transceiver.	
Diagram	N/A	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.
	ResultPtr	Pointer to the location where the cable diagnostics result shall be stored.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.19 Std_ReturnType EthTrcv_GetPhyIdentifier(uint8 TrcvIdx, uint32* OrgUniqueIdPtr, uint8* ModelNrPtr, uint8* RevisionNrPtr)

Function	Std_ReturnType EthTrcv_GetPhyIdentifier(uint8 TrcvIdx, uint32* OrgUniqueIdPtr, uint8* ModelNrPtr, uint8* RevisionNrPtr)	
Description	Obtains the PHY identifier of the Ethernet Transceiver.	
Diagram	<pre> graph LR A[EthTrcv_GetPhyIdentifier] --> B[EthTrcv_EcucPartitionCheck] A --> C[EthTrcv_GetPhyIdentifier_DevCheck] A --> D[EthTrcv_Hal_GetPhyIdentifier] D --> E[EthTrcv_Hw_ReadPhyId1] D --> F[EthTrcv_Hw_ReadPhyId2] </pre>	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.
	OrgUniqueIdPtr	Pointer to the memory where the Organizationally Unique Identifier shall be stored.
	ModelNrPtr	Pointer to the memory where the Manufacturer's Model Number shall be stored.
	RevisionNrPtr	Pointer to the memory where the Revision Number shall be stored.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.20 Std_ReturnType EthTrcv_GetMacMethod(uint8 TrcvIdx, EthTrcv_MacMethodType* MacModePtr)

Function	Std_ReturnType EthTrcv_GetMacMethod(uint8 TrcvIdx, EthTrcv_MacMethodType* MacModePtr)
Description	Obtains the media access mode of the transceiver when EthTrcvDuplexMode is configured as ETHTRCV_DUPLEX_MODE_HALF.

Diagram	<pre>graph LR; A[EthTrcv_GetMacMethod] --> B[EthTrcv_EcucPartitionCheck]; A --> C[EthTrcv_Hal_GetMacMethod]; A --> D[EthTrcv_StateIdxPtrDevCheck];</pre>							
Parameters	<table><tr><th>Parameter</th><th>Description</th></tr><tr><td>TrcvIdx</td><td>Index of the transceiver within the context of the Ethernet Transceiver Driver.</td></tr><tr><td>MacModePtr</td><td>ETHTRCV_MAC_TYPE_CSMA_CD: Carrier-sense multiple access with collision detection. ETHTRCV_MAC_TYPE_PLCA: Physical layer collision avoidance.</td></tr></table>	Parameter	Description	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.	MacModePtr	ETHTRCV_MAC_TYPE_CSMA_CD: Carrier-sense multiple access with collision detection. ETHTRCV_MAC_TYPE_PLCA: Physical layer collision avoidance.	
Parameter	Description							
TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.							
MacModePtr	ETHTRCV_MAC_TYPE_CSMA_CD: Carrier-sense multiple access with collision detection. ETHTRCV_MAC_TYPE_PLCA: Physical layer collision avoidance.							
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.							

2.7.1.21 void EthTrcv_GetVersionInfo(Std_VersionInfoType* VersionInfoPtr)

Function	void EthTrcv_GetVersionInfo(Std_VersionInfoType* VersionInfoPtr)	
Description	Returns the version information of this module.	
Diagram	N/A	
Parameters	Parameter	Description
	VersionInfoPtr	Version information of this module.
Returns	N/A	

2.7.1.22 void EthTrcv_MainFunction(void)

Function	void EthTrcv_MainFunction(void)		
Description	Ethernet Transceiver scheduled function.		
Diagram	<pre>graph LR; A[EthTrcv_MainFunction] --> B[EthTrcv_EcucPartitionCheck]; A --> C[EthTrcv_Hal_MainFunction]; C --> D[EthTrcv_Hw_GetTransceiverMode]</pre>		
Parameters	Parameter	Description	
Returns	N/A		

2.7.1.23 void EthTrcv_ReadMiiIndication(uint8 CtrlIdx,uint8 TrcvIdx,uint8 RegIdx,uint16 RegVal)

Function	void EthTrcv_ReadMiiIndication(uint8 CtrlIdx,uint8 TrcvIdx,uint8 RegIdx,uint16 RegVal)	
Description	Mii bus read indication.Called when information has been read out via MII interface.	
Diagram	N/A	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	TrcvIdx	Index of the transceiver on the MII.
	RegIdx	Index of the transceiver register on the MII.
	RegVal	Value contained in the indexed register.
Returns	N/A	

2.7.1.24 void EthTrcv_WriteMiiIndication(uint8 CtrlIdx,uint8 TrcvIdx,uint8 RegIdx)

Function	void EthTrcv_WriteMiiIndication(uint8 CtrlIdx,uint8 TrcvIdx,uint8 RegIdx)	
Description	Mii bus writeindication.Called when information has been written via MII interface.	
Diagram	N/A	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	TrcvIdx	Index of the transceiver on the MII.
	RegIdx	Index of the transceiver register on the MII.
Returns	N/A	

2.7.2 Local Functions in EthTrcv.c**2.7.2.1 LOCAL_INLINE boolean EthTrcv_StateIdxDevCheck(uint8 CtrlIdx, uint8 CoreId,uint8 Sid)**

Function	LOCAL_INLINE boolean EthTrcv_StateIdxDevCheck(uint8 CtrlIdx, uint8 CoreId,uint8 Sid)	
Description	Check the development errors of state and TrcvIdx.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.
	CoreId	Partition ID of current processor.
	Sid	Service ID of the caller.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	EthTrcv_SetTransceiverMode(),EthTrcv_SetTransceiverWakeupMode(),EthTrcv_CheckWakeup(),EthTrcv_StartAutoNegotiation(),EthTrcv_TransceiverLinkStateRequest(),EthTrcv_SetPhyTestMode(),EthTrcv_SetPhyLoopbackMode(), EthTrcv_SetPhyTxMode(), EthTrcv_RunCableDiagnostic()	

2.7.2.2 LOCAL_INLINE boolean EthTrcv_StateIdxPtrDevCheck (uint8 CtrlIdx, uint8 CoreId, const void* Ptr,uint8 Sid)

Function	LOCAL_INLINE boolean EthTrcv_StateIdxDevCheck(uint8 CtrlIdx, uint8 CoreId,uint8 Sid)	
Description	Check the development errors of state and TrcvIdx.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.
	CoreId	Partition ID of current processor.
	Ptr	Pointer to memory.
	Sid	Service ID of the caller.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	EthTrcv_GetTransceiverMode(),EthTrcv_GetTransceiverWakeupMode(),EthTrcv_GetBusWuReason(),EthTrcv_GetLinkState(),EthTrcv_GetBaudRate(),EthTrcv_GetDuplexMode(),EthTrcv_GetPhySignalQuality(),EthTrcv_GetCableDiagnosticsResult(),EthTrcv_SetPhyTxMode,EthTrcv_RunCableDiagnostic(),EthTrcv_GetMacMethod()	

2.7.2.3 LOCAL_INLINE boolean EthTrcv_InitDevCheck(const EthTrcv_ConfigType* CfgPtr)

Function	LOCAL_INLINE boolean EthTrcv_InitDevCheck(const EthTrcv_ConfigType* CfgPtr)
Description	Check the development errors of EthTrcv_Init.

Parameters	Parameter	Description
	CfgPtr	Points to the implementation specific structure.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	EthTrcv_Init	

2.7.2.4 LOCAL_INLINE boolean EthTrcv_GetPhyIdentifierDevCheck(uint8 TrcvIdx,uint8 CoreId,uint32* OrgUniqueldPtr,uint8* ModelNrPtr,uint8* RevisionNrPtr)

Function	LOCAL_INLINE boolean EthTrcv_GetPhyIdentifierDevCheck(uint8 TrcvIdx,uint8 CoreId,uint32* OrgUniqueldPtr,uint8* ModelNrPtr,uint8* RevisionNrPtr)	
Description	Check the development errors of Eth_GetControllerMode.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver.
	CoreId	Partition ID of current processor.
	OrgUniqueldPtr	Pointer to the memory where the Organizationally Unique Identifier shall be stored.
	ModelNrPtr	Pointer to the memory where the Manufacturer's Model Number shall be stored.
	RevisionNrPtr	Pointer to the memory where the Revision Number shall be stored.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_GetControllerMode ().	

2.7.2.5 LOCAL_INLINE boolean EthTrcv_EcucPartitionCheck(const EthTrcv_TrcvCfgType *TrcvCfgPtr,uint8 CoreId,uint8 Sid)

Function	LOCAL_INLINE boolean EthTrcv_EcucPartitionCheck(const EthTrcv_TrcvCfgType *TrcvCfgPtr,uint8 CoreId,uint8 Sid)	
Description	Check the ECUC partition.	
Parameters	Parameter	Description
	TrcvCfgPtr	Pointer to the controller configuration.
	CoreId	Partition ID of current processor.
	Sid	Service ID of the caller.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By		

2.8 Hardware Functions

2.8.1 Functions in EthTrcv_Hal.h

2.8.1.1 Std_ReturnType EthTrcv_Hal_CheckAccessTransceiver(const EthTrcv_TrcvCfgType *TrcvCfgPtr)

Function	Std_ReturnType EthTrcv_Hal_CheckAccessTransceiver(const EthTrcv_TrcvCfgType *TrcvCfgPtr)	
Description	Check whether the transceiver is accessible.	
Parameters	Parameter	Description
	TrcvCfgPtr	Pointer to the transceiver configuration.
Returns	E_OK: The controller is accessible.	

	E_NOT_OK: The controller is not accessible.
Referenced By	EthTrcv_Init ().

2.8.1.2 Std_ReturnType EthTrcv_Hal_InitTransceiver(uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr)

Function	Std_ReturnType EthTrcv_Hal_InitTransceiver(uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr)	
Description	Initialize the indexed transceiver.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
Returns	The index of TX FIFO.	
Referenced By	EthTrcv_Init ()	

2.8.1.3 Std_ReturnType EthTrcv_Hal_SetTransceiverMode(uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr,Eth_ModeType TrcvMode)

Function	Std_ReturnType EthTrcv_Hal_SetTransceiverMode(uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr,Eth_ModeType TrcvMode)	
Description	Set the transceiver mode of indexed transceiver.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
	TrcvMode	ETHTRCV_MODE_DOWN: disable the transceiver. ETHTRCV_MODE_ACTIVE: enable the transceiver. ETH_MODE_ACTIVE_WITH_WAKEUP_REQUEST: enable the transceiver and request an wake-up on the network
Returns	E_OK The transceiver mode is set successfully. E_NOT_OK The transceiver mode is set failed.	
Referenced By	EthTrcv_SetTransceiverMode ()	

2.8.1.4 Std_ReturnType EthTrcv_Hal_GetTransceiverMode(uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr,Eth_ModeType TrcvModePtr)

Function	Std_ReturnType EthTrcv_Hal_GetTransceiverMode(uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr,Eth_ModeType *TrcvModePtr)	
Description	Get the transceiver mode of indexed transceiver.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
	TrcvModePtr	ETHTRCV_MODE_DOWN: disable the transceiver. ETHTRCV_MODE_ACTIVE: enable the transceiver.
Returns	E_OK Get the transceiver mode successfully. E_NOT_OK Get the transceiver mode failed.	
Referenced By	EthTrcv_GetTransceiverMode ()	

2.8.1.5 Std_ReturnType EthTrcv_Hal_SetTransceiver WakeupMode(uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_WakeupModeType TrcvWakeupMode)

Function	Std_ReturnType EthTrcv_Hal_SetTransceiverMode(uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_WakeupModeType TrcvWakeupMode)	
Description	Set the transceiver wake up mode of indexed transceiver.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
	TrcvWakeupMode	ETHTRCV_WUM_DISABLE: disable transceiver wake up. ETHTRCV_WUM_ENABLE: enable transceiver wake up. ETHTRCV_WUM_CLEAR: clears transceiver wake up reason.
Returns	E_OK success. E_NOT_OK failed.	
Referenced By	EthTrcv_SetTransceiverWakeupMode ()	

2.8.1.6 Std_ReturnType EthTrcv_Hal_GetTransceiver WakeupMode(uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_WakeupModeType TrcvWakeupModePtr)

Function	Std_ReturnType EthTrcv_Hal_GetTransceiverMode(uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr,Eth_ModeType * TrcvWakeupModePtr)	
Description	Get the transceiver wake up mode of indexed transceiver.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
	TrcvWakeupModePtr	ETHTRCV_WUM_DISABLE: disable transceiver wake up. ETHTRCV_WUM_ENABLE: enable transceiver wake up.
Returns	E_OK success. E_NOT_OK failed.	
Referenced By	EthTrcv_GetTransceiverWakeupMode ()	

2.8.1.7 Std_ReturnType EthTrcv_Hal_GetBusWuReason(uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr,EthTrcv_WakeupReasonType* WakeupReasonPtr)

Function	Std_ReturnType EthTrcv_Hal_GetBusWuReason(uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr,EthTrcv_WakeupReasonType* WakeupReasonPtr)	
Description	Get the least recent wakeup reasons of indexed transceiver.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
	WakeupReasonPtr	The reasons defined in EthTrcv_WakeupReasonType.
Returns	E_OK success. E_NOT_OK failed.	
Referenced By	EthTrcv_GetBusWuReason ()	

2.8.1.8 Std_ReturnType EthTrcv_Hal_CheckWakeup (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr)

Function	Std_ReturnType EthTrcv_Hal_CheckWakeup (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr)	
Description	Check if the wakeup event of the indexed transceiver is detected.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
Returns	E_OK success. E_NOT_OK failed.	
Referenced By	EthTrcv_CheckWakeup ()	

2.8.1.9 Std_ReturnType EthTrcv_Hal_StartAutoNegotiation (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr)

Function	Std_ReturnType EthTrcv_Hal_StartAutoNegotiation (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr)	
Description	Restart the automatic negotiation of the transmission.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
Returns	E_OK success. E_NOT_OK failed.	
Referenced By	EthTrcv_StartAutoNegotiation ()	

2.8.1.10 Std_ReturnType EthTrcv_Hal_TransceiverLinkStateRequest (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_LinkStateType LinkState)

Function	Std_ReturnType EthTrcv_Hal_TransceiverLinkStateRequest (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_LinkStateType LinkState)	
Description	Start link training of the indexed transceiver.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
	LinkState	The Ethernet link state of a physical Ethernet connection.
Returns	E_OK success. E_NOT_OK failed.	
Referenced By	EthTrcv_TransceiverLinkStateRequest ()	

2.8.1.11 Std_ReturnType EthTrcv_Hal_GetLinkState (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_LinkStateType* LinkStatePtr)

Function	Std_ReturnType EthTrcv_Hal_GetLinkState (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_LinkStateType* LinkStatePtr)	
Description	Read the current link state of the indexed transceiver.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver

	TrcvCfgPtr	Pointer to the transceiver configuration.
	LinkStatePtr	ETHTRCV_LINK_STATE_DOWN: transceiver is disconnected. ETHTRCV_LINK_STATE_ACTIVE: transceiver is connected.
Returns	E_OK success. E_NOT_OK failed.	
Referenced By	EthTrcv_GetLinkState ()	

2.8.1.12 Std_ReturnType EthTrcv_Hal_GetBaudRate (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_BaudRateType* BaudRatePtr)

Function	Std_ReturnType EthTrcv_Hal_GetBaudRate (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_BaudRateType* BaudRatePtr)	
Description	Read the baud rate of the current link.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
	BaudRatePtr	ETHTRCV_BAUD_RATE_10MBIT: 10MBit connection. ETHTRCV_BAUD_RATE_100MBIT: 100MBit connection. ETHTRCV_BAUD_RATE_1000MBIT: 1000MBit connection. ETHTRCV_BAUD_RATE_2500MBIT: 2500MBIT Ethernet connection.
Returns	E_OK success. E_NOT_OK failed.	
Referenced By	EthTrcv_GetBaudRate ()	

2.8.1.13 Std_ReturnType EthTrcv_Hal_GetDuplexMode (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_DuplexModeType* DuplexModePtr)

Function	Std_ReturnType EthTrcv_Hal_GetDuplexMode (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_DuplexModeType* DuplexModePtr)	
Description	Read the duplex mode of the current link.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
	DuplexModePtr	ETHTRCV_DUPLEX_MODE_HALF: half duplex connections. ETHTRCV_DUPLEX_MODE_FULL: full duplex connection.
Returns	E_OK success. E_NOT_OK failed.	
Referenced By	EthTrcv_GetDuplexMode ()	

2.8.1.14 Std_ReturnType EthTrcv_Hal_SetPhyTestMode (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_PhyTestModeType Mode)

Function	Std_ReturnType EthTrcv_Hal_SetPhyTestMode (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_PhyTestModeType Mode)	
Description	Activates a given test mode of the indexed transceiver.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver

		Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
	Mode	Test mode to be activated.
Returns	E_OK success. E_NOT_OK failed.	
Referenced By	EthTrcv_SetPhyTestMode ()	

2.8.1.15 Std_ReturnType EthTrcv_Hal_SetPhyLoopbackMode (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_PhyLoopbackModeType Mode)

Function	Std_ReturnType EthTrcv_Hal_SetPhyLoopbackMode (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_PhyLoopbackModeType Mode)	
Description	Activates a given loopback mode of the indexed transceiver.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
	Mode	Loopback mode to be activated.
Returns	E_OK success. E_NOT_OK failed.	
Referenced By	EthTrcv_SetPhyLoopbackMode ()	

2.8.1.16 Std_ReturnType EthTrcv_Hal_GetPhySignalQuality (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, uint32* SignalQualityPtr)

Function	Std_ReturnType EthTrcv_Hal_GetPhySignalQuality (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, uint32* SignalQualityPtr)	
Description	Read the signal quality of the current link.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
	SignalQualityPtr	Pointer to the memory where the signal quality shall be stored.
Returns	E_OK success. E_NOT_OK failed.	
Referenced By	EthTrcv_GetPhySignalQuality ()	

2.8.1.17 Std_ReturnType EthTrcv_Hal_SetPhyTxMode (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_PhyTxModeType Mode)

Function	Std_ReturnType EthTrcv_Hal_SetPhyTxMode (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_PhyTxModeType Mode)	
Description	Set the transmission mode of the indexed transceiver.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
	Mode	Transmission mode to be activated.
Returns	E_OK success.	

	E_NOT_OK failed.
Referenced By	EthTrcv_SetPhyTxMode ()

2.8.1.18 Std_ReturnType EthTrcv_Hal_RunCableDiagnostic (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr)

Function	Std_ReturnType EthTrcv_Hal_RunCableDiagnostic (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr)	
Description	Trigger the cable diagnostics of the indexed transceiver.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
Returns	E_OK success. E_NOT_OK failed.	
Referenced By	EthTrcv_RunCableDiagnostic ()	

2.8.1.19 Std_ReturnType EthTrcv_Hal_GetCableDiagnosticsResult (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_CableDiagResultType* ResultPtr)

Function	Std_ReturnType EthTrcv_Hal_GetCableDiagnosticsResult (uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr, EthTrcv_CableDiagResultType* ResultPtr)	
Description	Get the cable diagnostics result of the indexed transceiver.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
	ResultPtr	Pointer to the location where the cable diagnostics result shall be stored.
Returns	E_OK success. E_NOT_OK failed.	
Referenced By	EthTrcv_GetCableDiagnosticsResult ()	

2.8.1.20 Std_ReturnType EthTrcv_Hal_GetPhyIdentifier(const EthTrcv_TrcvCfgType *TrcvCfgPtr,uint32* OrgUniqueldPtr,uint8* ModelNrPtr,uint8* RevisionNrPtr)

Function	Std_ReturnType EthTrcv_Hal_GetPhyIdentifier(const EthTrcv_TrcvCfgType *TrcvCfgPtr,uint32* OrgUniqueldPtr,uint8* ModelNrPtr,uint8* RevisionNrPtr)	
Description	Read the identifier of the indexed transceiver.	
Parameters	Parameter	Description
	TrcvCfgPtr	Pointer to the transceiver configuration.
	OrgUniqueldPtr	Pointer to the memory where the Organizationally Unique Identifier shall be stored.
	ModelNrPtr	Pointer to the memory where the Manufacturer's Model Number shall be stored.
	RevisionNrPtr	Pointer to the memory where the Revision Number shall be stored.
Returns	E_OK success. E_NOT_OK failed.	
Referenced By	EthTrcv_GetPhyIdentifier ()	

2.8.1.21 Std_ReturnType EthTrcv_Hal_GetMacMethod(uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr,EthTrcv_MacMethodType* MacModePtr)

Function	Std_ReturnType EthTrcv_Hal_GetMacMethod(uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr,EthTrcv_MacMethodType* MacModePtr)	
Description	Get the media access mode of the transceiver when EthTrcvDuplexMode is configured as ETHTRCV_DUPLEX_MODE_HALF.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
	MacModePtr	ETHTRCV_MAC_TYPE_CSMA_CD: Carrier-sense multiple access with collision detection. ETHTRCV_MAC_TYPE_PLCA: Physical layer collision avoidance.
Returns	E_OK success. E_NOT_OK failed.	
Referenced By	EthTrcv_GetMacMethod ()	

2.8.1.22 void EthTrcv_Hal_MainFunction(uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr)

Function	void EthTrcv_Hal_MainFunction(uint8 TrcvIdx,const EthTrcv_TrcvCfgType *TrcvCfgPtr)	
Description	Ethernet Transceiver scheduled function. Used for polling state changes and wakeup reasons. Calls EthIf_TrcvModeIndication when the transceiver mode changed. Stores wakeup events if EthTrcvWakeUpSupport is set to ETHTRCV_WAKEUP_BY_POLLING.	
Parameters	Parameter	Description
	TrcvIdx	Index of the transceiver within the context of the Ethernet Transceiver Driver
	TrcvCfgPtr	Pointer to the transceiver configuration.
Returns	N/A	
Referenced By	EthTrcv_MainFunction ()	

2.8.2 Functions in EthTrcv_Hw.h

2.8.2.1 Std_ReturnType EthTrcv_Hw_ReadPhyId1(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,uint16 *RegValPtr)

Function	Std_ReturnType EthTrcv_Hw_ReadPhyId1(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,uint16 *RegValPtr)	
Description	Read the PHY identifier 1 register.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	RegValPtr	Pointer to the address to save the register value.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.2 Std_ReturnType EthTrcv_Hw_ReadPhyId2(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,uint16 *RegValPtr)

Function	Std_ReturnType EthTrcv_Hw_ReadPhyId2(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,uint16 *RegValPtr)	
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Description	Read the PHY identifier 2 register.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	RegValPtr	Pointer to the address to save the register value.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.3 Std_ReturnType EthTrcv_Hw_SoftReset(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr)

Function	Std_ReturnType EthTrcv_Hw_SoftReset(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr)	
Description	Perform software reset.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.4 Std_ReturnType EthTrcv_Hw_ReadResetFlag (EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr, boolean *ResetBusyPtr)

Function	Std_ReturnType EthTrcv_Hw_ReadResetFlag (EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr, boolean *ResetBusyPtr)	
Description	Read software reset busy flag.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	ResetBusyPtr	Pointer to the memory to save the soft reset busy state.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.5 Std_ReturnType EthTrcv_Hw_SetRgmiiTxDelay(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,uint8 DelayValue)

Function	Std_ReturnType EthTrcv_Hw_SetRgmiiTxDelay(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,uint8 DelayValue)	
Description	Set RGMII TX clock delay.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	DelayValue	TX clock delay value.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.6 Std_ReturnType EthTrcv_Hw_SetRgmiiRxDelay(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,uint8 DelayValue)

Function	Std_ReturnType EthTrcv_Hw_SetRgmiiRxDelay(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,uint8 DelayValue)	
Description	Set RGMII RX clock delay.	

Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	DelayValue	RX clock delay value.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.7 Std_ReturnType EthTrcv_Hw_SpecificConfig(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,EthTrcv_BaudRateType Speed,EthTrcv_DuplexModeType Duplex)

Function	Std_ReturnType EthTrcv_Hw_SpecificConfig(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,EthTrcv_BaudRateType Speed,EthTrcv_DuplexModeType Duplex)	
Description	Hardware specific configuration.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	Speed	Baud Rate from configuration structure.
	Duplex	Duplex mode from configuration structure.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.8 Std_ReturnType EthTrcv_Hw_BridgeConfig(EthTrcv_DerivateType Derivate, const EthTrcv_MiilfType *MiilfPtr, EthTrcv_BaudRateType Speed, EthTrcv_DuplexModeType Duplex)

Function	Std_ReturnType EthTrcv_Hw_BridgeConfig(EthTrcv_DerivateType Derivate, const EthTrcv_MiilfType *MiilfPtr, EthTrcv_BaudRateType Speed, EthTrcv_DuplexModeType Duplex)	
Description	Hardware specific configuration.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	Speed	Baud Rate from configuration structure.
	Duplex	Duplex mode from configuration structure.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.9 Std_ReturnType EthTrcv_Hw_SetClockOutput(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,EthTrcv_ClkOutPutType ClkOutMode)

Function	Std_ReturnType EthTrcv_Hw_SetClockOutput(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,EthTrcv_ClkOutPutType ClkOutMode)	
Description	Set clock output mode.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	ClkOutMode	Clock output mode.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.10 Std_ReturnType EthTrcv_Hw_EnableAutoNegtiation (EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr, boolean EnableAutoNeg)

Function	Std_ReturnType EthTrcv_Hw_EnableAutoNegtiation (EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr, boolean EnableAutoNeg)	
Description	Enable or distable Auto-Negotiation.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	EnableAutoNeg	TRUE: Enable, FALSE: Disable.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.11 Std_ReturnType EthTrcv_Hw_ReadAutoNegtiationStatus (EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr, boolean *CompleteStatusPtr)

Function	Std_ReturnType EthTrcv_Hw_ReadAutoNegtiationStatus (EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr, boolean *CompleteStatusPtr)	
Description	Read Auto-Negotiation completed status.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	CompleteStatusPtr	Pointer to the memory to save the Auto-Negotiation completed status.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.12 Std_ReturnType EthTrcv_Hw_SetAutoNegtiationAbility(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,EthTrcv_BaudRateType MaxSpeed)

Function	Std_ReturnType EthTrcv_Hw_SetAutoNegtiationAbility(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,EthTrcv_BaudRateType MaxSpeed)	
Description	Set Auto-Negotiation advertise ability.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	MaxSpeed	The maximum speed advertised for Auto-Negotiation.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.13 Std_ReturnType EthTrcv_Hw_ManualLink(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,EthTrcv_DuplexModeType DuplexMode,EthTrcv_BaudRateType SpeedMode)

Function	Std_ReturnType EthTrcv_Hw_ManualLink(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,EthTrcv_DuplexModeType DuplexMode,EthTrcv_BaudRateType SpeedMode)	
Description	Set manual duplex and speed selection.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	DuplexMode	Duplex mode from configuration structure.

	SpeedMode	Baud Rate from configuration structure.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.14 Std_ReturnType EthTrcv_Hw_ReadLinkState(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,EthTrcv_LinkStateType* LinkStatePtr)

Function	Std_ReturnType EthTrcv_Hw_ReadLinkState(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,EthTrcv_LinkStateType* LinkStatePtr)	
Description	Read the current link state of the transceiver.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	LinkStatePtr	Pointer to the address to save the link state.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.15 Std_ReturnType EthTrcv_Hw_ReadBaudRate (EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr, EthTrcv_BaudRateType* BaudRatePtr)

Function	Std_ReturnType EthTrcv_Hw_ReadBaudRate (EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr, EthTrcv_BaudRateType* BaudRatePtr)	
Description	Read the current baud rate of the transceiver.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	BaudRatePtr	Pointer to the address to save the baud rate.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.16 Std_ReturnType EthTrcv_Hw_ReadDuplexMode (EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr, EthTrcv_DuplexModeType* DuplexModePtr)

Function	Std_ReturnType EthTrcv_Hw_ReadDuplexMode (EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr, EthTrcv_DuplexModeType* DuplexModePtr)	
Description	Read the current duplex mode of the transceiver.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	DuplexModePtr	Pointer to the address to save the duplex mode.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.17 Std_ReturnType EthTrcv_Hw_GetTransceiverMode(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,Eth_ModeType* TrcvModePtr)

Function	Std_ReturnType EthTrcv_Hw_GetTransceiverMode(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,Eth_ModeType* TrcvModePtr)	
Description	Get the current mode of the transceiver.	
Parameters	Parameter	Description

	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	TrcvModePtr	ETHTRCV_MODE_DOWN: the transceiver is disabled. ETHTRCV_MODE_ACTIVE: the transceiver is enabled.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.18 Std_ReturnType EthTrcv_Hw_SetTransceiverUpDown (EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr, boolean PowerUp)

Function	Std_ReturnType EthTrcv_Hw_SetTransceiverUp (EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr, boolean PowerUp)	
Description	Set transceiver mode up.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	PowerUp	Set transceiver mode up or down
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.19 boolean EthTrcv_Hw_TestModeSupported(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,EthTrcv_PhyTestModeType Mode)

Function	boolean EthTrcv_Hw_TestModeSupported(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,EthTrcv_PhyTestModeType Mode)	
Description	Check the test mode is supported by the transceiver.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	Mode	Test mode.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.20 Std_ReturnType EthTrcv_Hw_SetTestMode(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,EthTrcv_PhyTestModeType Mode)

Function	Std_ReturnType EthTrcv_Hw_SetTestMode(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,EthTrcv_PhyTestModeType Mode)	
Description	Set PHY test mode.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	Mode	Test mode.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.21 Std_ReturnType EthTrcv_Hw_SetLoopbackMode(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,EthTrcv_PhyLoopbackModeType Mode)

Function	Std_ReturnType EthTrcv_Hw_SetLoopbackMode(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType	
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	*MiilfPtr,EthTrcv_PhyLoopbackModeType Mode)	
Description	Set PHY loopback mode.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	Mode	Loopback mode.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.22 Std_ReturnType EthTrcv_Hw_SignalQualitySupported (EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr)

Function	Std_ReturnType EthTrcv_Hw_SignalQualitySupported (EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr)	
Description	Check the Signal Quality is supported by the transceiver.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
Returns	E_OK success. E_NOT_OK failed.	

2.8.2.23 Std_ReturnType EthTrcv_Hw_GetSignalQuality(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,uint32* SignalQualityPtr)

Function	Std_ReturnType EthTrcv_Hw_GetSignalQuality(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,uint32* SignalQualityPtr)	
Description	Get the Signal Quality of current link.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	SignalQualityPtr	Pointer to the memory where the signal quality shall be stored.
Returns	E_OK success. E_NOT_OK failed.	

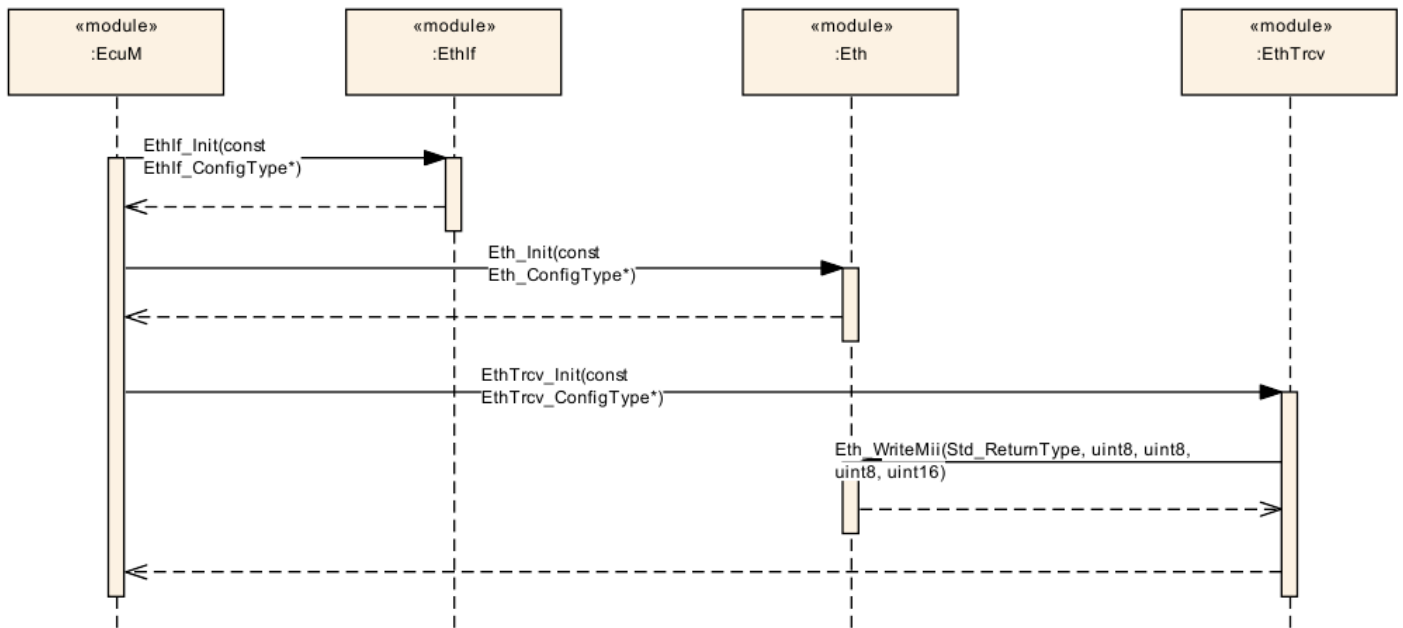
2.8.2.24 Std_ReturnType EthTrcv_Hw_SetPhyTxMode(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,EthTrcv_PhyTxModeType Mode)

Function	Std_ReturnType EthTrcv_Hw_SetPhyTxMode(EthTrcv_DerivateType Derivate,const EthTrcv_MiilfType *MiilfPtr,EthTrcv_PhyTxModeType Mode)	
Description	Set PHY transmission mode.	
Parameters	Parameter	Description
	Derivate	Derivate type of the hardware configuration.
	MiilfPtr	Pointer to the MII interface structure.
	Mode	Transmission mode.
Returns	E_OK success. E_NOT_OK failed.	

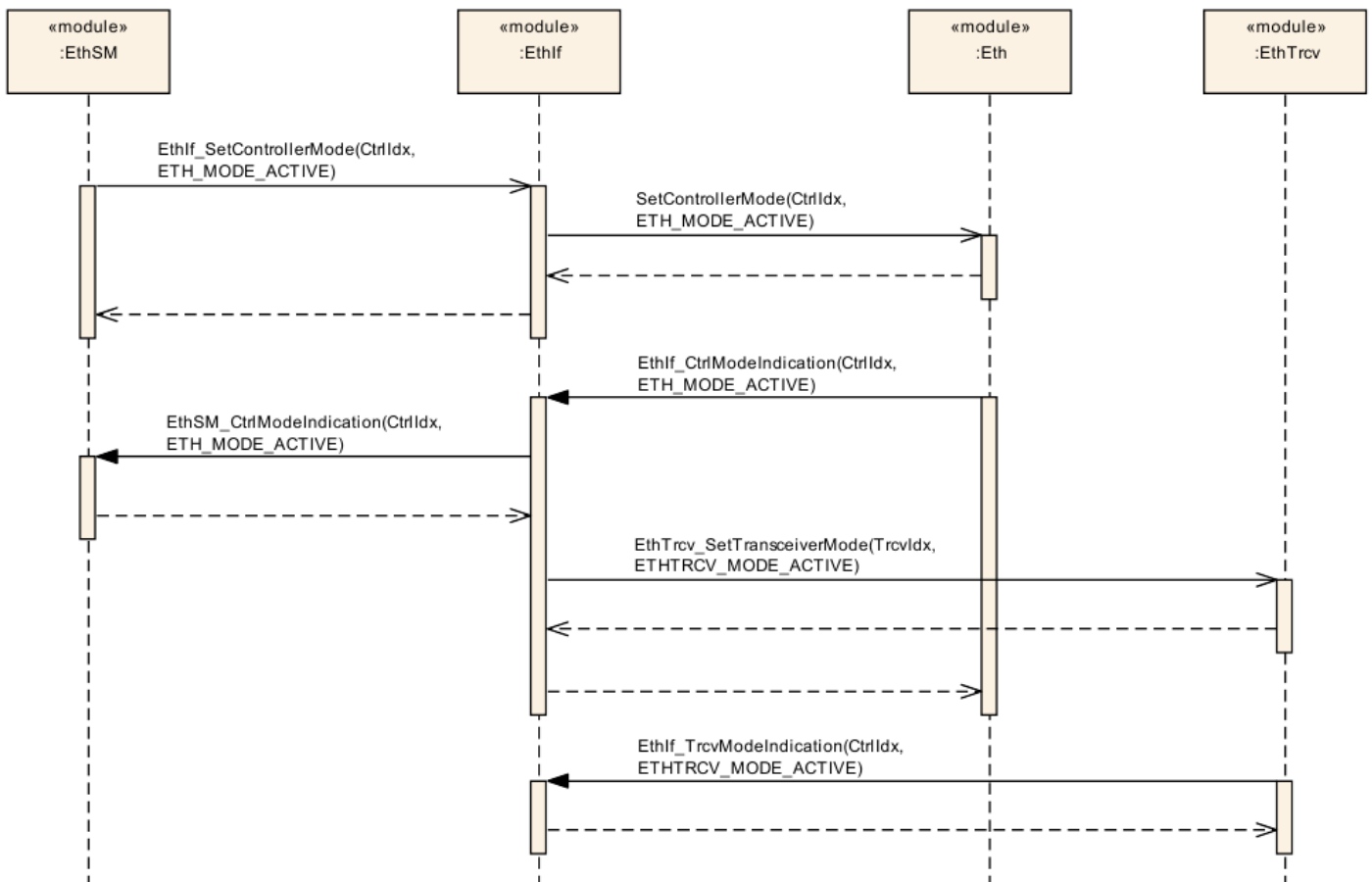
2.9 API Sequence Diagram

2.9.1 Initialization

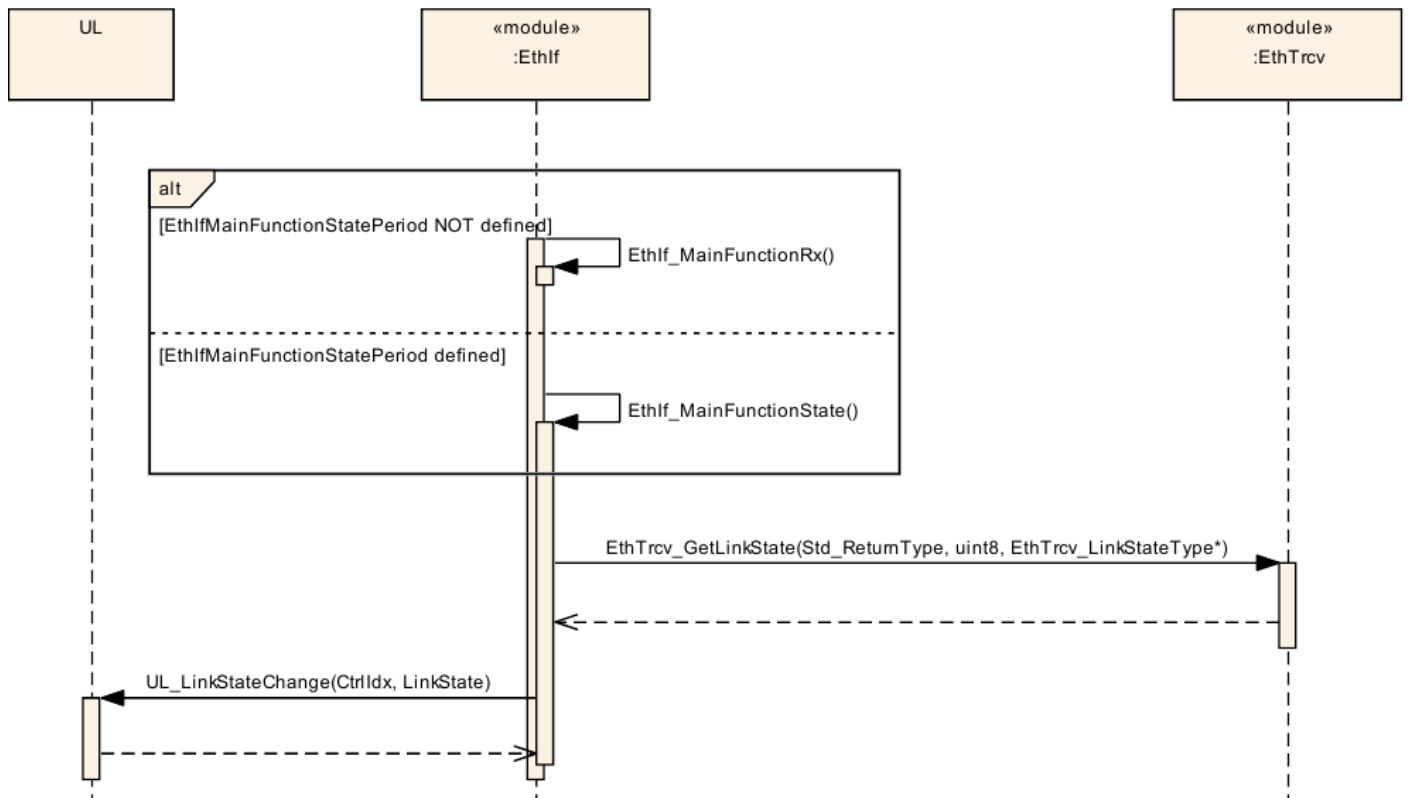
The ECU State Manager (EcuM) is responsible for calling the initialization function.



2.9.2 Communication Initialization



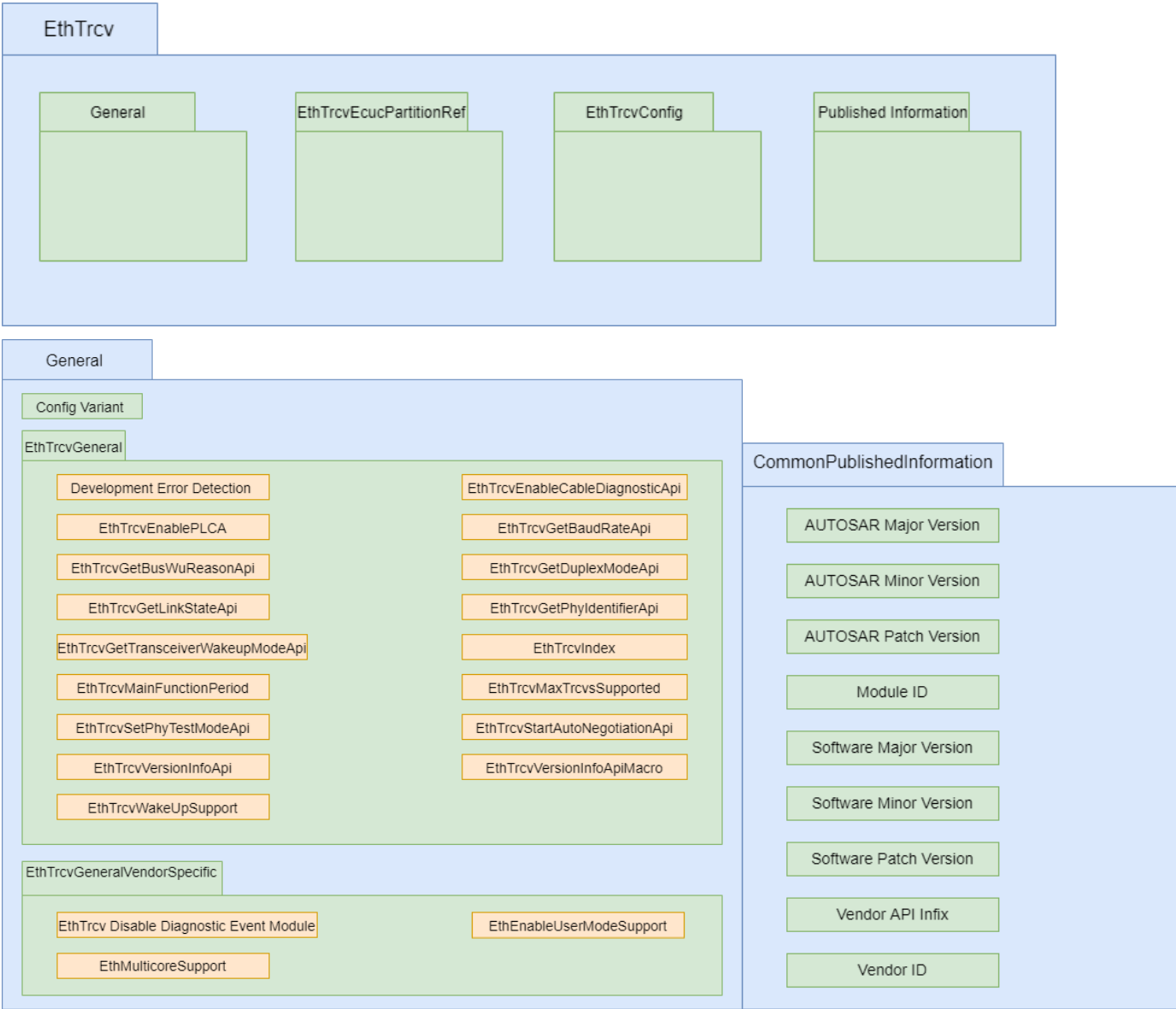
2.9.3 Link State Change

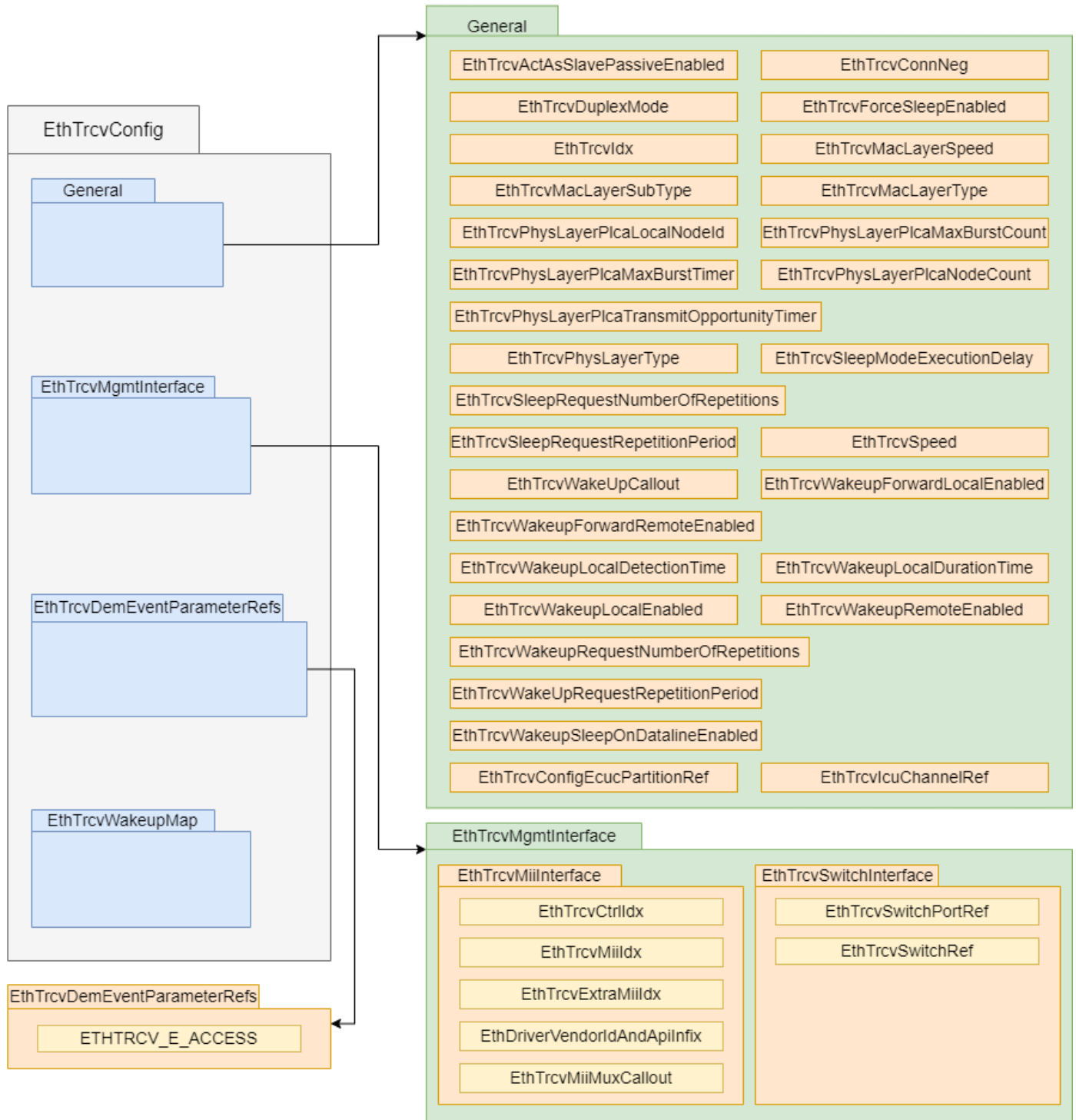


Chapter 3 Tresos Configuration Items

3.1 Container Inclusion Relation

The container inclusion relation is shown as below:





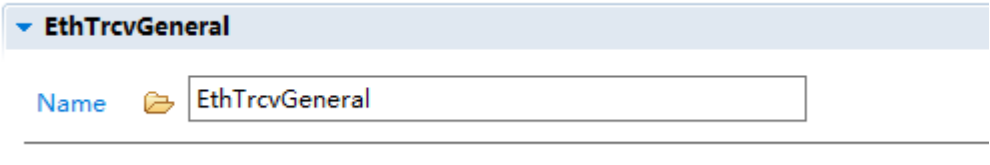
3.2 Containers and Variables

3.2.1 IMPLEMENTATION_CONFIG_VARIANT

Container	IMPLEMENTATION_CONFIG_VARIANT	
Description	N/A	
Screenshot	Config Variant VariantPreCompile	
Properties	Property	Value
	Type	Variable: enumeration
	Label	Config Variant

	Default	VariantPostBuild
	Range	VariantPostBuild VariantPreCompile
Returns	N/A	

3.2.2 EthTrcvGeneral

Container	EthTrcvGeneral	
Description	General configuration of Ethernet Transceiver Driver module.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.2.1 Development Error Detection

Container	Development Error Detection	
Description	Switches the development error detection and notification on or off.	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicName	False
	Default	False

3.2.2.2 EthTrcvEnableCableDiagnosticApi

Container	EthTrcvEnableCableDiagnosticApi	
Description	Enable/disable the APIs for cable diagnostic: EthTrcv_RunCableDiagnostic, EthTrcv_GetCableDiagnosticsResult	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicName	False
	Default	False

3.2.2.3 EthEnablePLCA

Container	EthEnablePLCA	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	Boolean

	Origin	AUTOSAR_ECUC
	SynbolicName	False
	Default	False

3.2.2.4 EthTrcvGetBaudRateApi

Container	EthTrcvGetBaudRateApi	
Description	Enables / Disables EthTrcv_GetBaudRate API.	
Screenshot	EthTrcvGetBaudRateApi   	
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SynbolicName	False
	Default	True

3.2.2.5 EthTrcvGetBusWuReasonApi

Variable	EthTrcvGetBusWuReasonApi	
Description	Enables / Disables API to obtain the recent wake-up reason detected by the used Ethernet hardware (e.g. PHY).	
Screenshot	EthTrcvGetBusWuReasonApi   	
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.2.6 EthTrcvGetDuplexModeApi

Variable	EthTrcvGetDuplexModeApi	
Description	Enables / Disables EthTrcv_GetDuplexMode API	
Screenshot	EthTrcvGetDuplexModeApi   	
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	True

3.2.2.7 EthTrcvGetLinkStateApi

Variable	EthTrcvGetLinkStateApi	
Description	Enables / Disables EthTrcv_GetLinkState API	
Screenshot	EthTrcvGetLinkStateApi   	
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC

	SymbolicNameValue	False
	Default	True

3.2.2.8 EthTrcvGetPhyIdentifierApi

Variable	EthTrcvGetPhyIdentifierApi	
Description	Enables / Disables EthTrcv_GetPhyIdentifier API.	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	True

3.2.2.9 EthTrcvGetTransceiverWakeupModeApi

Variable	EthTrcvGetTransceiverWakeupModeApi	
Description	Enables / Disables EthTrcv_GetTransceiverWakeupMode API	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.2.10 EthTrcvIndex

Variable	EthTrcvIndex	
Description	Specifies the InstanceId of this module instance. If only one instance is present it shall have the Id 0.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	0

3.2.2.11 EthTrcvMainFunctionPeriod

Variable	EthTrcvMainFunctionPeriod	
Description	Specifies the period of main function EthTrcv_MainFunction in seconds.	
Screenshot		
Properties	Property	Value
	Type	Variable: Float
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	0.001

3.2.2.12 EthTrcvMaxTrcvsSupported

Variable	EthTrcvMaxTrcvsSupported	
Description	Limits the total number of supported transceivers.	
Screenshot	EthTrcvMaxTrcvsSupported (1 -> 255)  1 	
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	1

3.2.2.13 EthTrcvSetPhyTestModeApi

Variable	EthTrcvSetPhyTestModeApi	
Description	Enables / Disables EthTrcv_SetPhyTestMode API.	
Screenshot	EthTrcvSetPhyTestModeApi  ☐ 	
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.2.14 EthTrcvStartAutoNegotiationApi

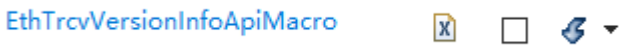
Container	EthTrcvStartAutoNegotiationApi	
Description	Enables / Disables EthTrcv_StartAutoNegotiation API	
Screenshot	EthTrcvStartAutoNegotiationApi  ☒ 	
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	True

3.2.2.15 EthTrcvVersionInfoApi

Variable	EthTrcvVersionInfoApi	
Description	Enables / Disables version info API	
Screenshot	EthTrcvVersionInfoApi  ☒ 	
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	True

3.2.2.16 EthTrcvVersionInfoApiMacro

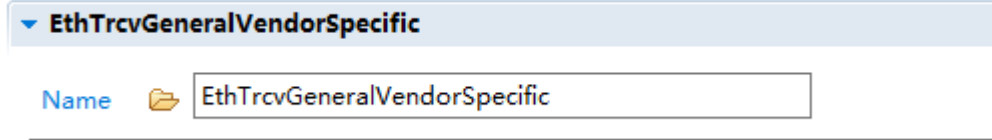
Variable	EthTrcvVersionInfoApiMacro
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Description	Enables / Disables version info API macro implementation	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.2.17 EthTrcvWakeUpSupport

Variable	EthTrcvWakeUpSupport	
Description	Configures how to detect a signaled wake-up by hardware: polling, asynchronous check via signal, interrupt or to not used/not supported.	
Screenshot		
Properties	Property	Value
	Type	Enumeration
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Range	ETHTRCV_WAKEUP_BY_ ASYNCHRONOUS_CHECK Wake-up detection is done within the main function. Polling of the Ethernet hardware is only done, if a wake-up was signaled. Otherwise the Ethernet hardware is not checked for wake-up.Used e.g. if Ethernet hardware is OA TC10 compliant and the ECU maintain an Ethernet switch. ETHTRCV_WAKEUP_BY_INTERRUPT Wake-up detection is signaled by interrupt ETHTRCV_WAKEUP_BY_POLLING Wake-up detection is done by polling continuously within the main function ETHTRCV_WAKEUP_NOT_SUPPORTED Wake up is not supported
	Default	ETHTRCV_WAKEUP_NOT_SUPPORTED

3.2.2.18 EthTrcvGeneralVendorSpecific

Container	EthTrcvGeneralVendorSpecific	
Description	Vendor specific EthTrcv general configurations.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.2.18.1 EthTrcv Disable Diagnostic Event Module

Variable	EthTrcv Disable Diagnostic Event Module	
Description	Disables / Enables EthTrcv Diagnostic Event Module .	
Screenshot		

Properties	Property	Value
	Type	Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

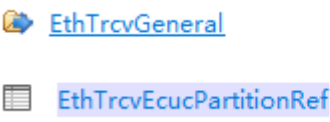
3.2.2.18.2 EthTrcvEnableUserModeSupport

Variable	EthTrcvEnableUserModeSupport	
Description	Enables / Disables user mode .	
Screenshot	EthTrcvEnableUserModeSupport 	
Properties	Property	Value
	Type	Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

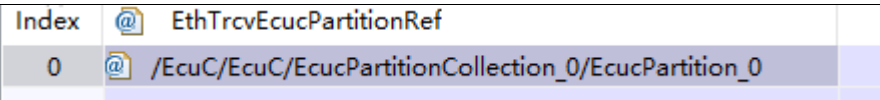
3.2.2.18.3 EthTrcvMulticoreSupport

Variable	EthTrcvMulticoreSupport	
Description	Enables / Disables multiple core partitions.	
Screenshot	EthTrcvMulticoreSupport 	
Properties	Property	Value
	Type	Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.2.19 EthTrcvEcucPartitionRef

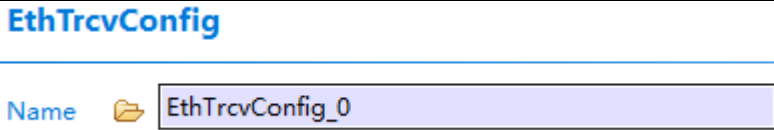
Container	EthTrcvEcucPartitionRef	
Description	Maps the Ethernet transceiver driver to zero or multiple ECUC partitions to make the modules API available in this partition. The Ethernet transceiver driver will operate as an independent instance in each of the partitions.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.2.19.1 EthTrcvEcucPartitionRef

Variable	EthTrcvEcucPartitionRef	
Description	Reference to an ECUC partition.	
Screenshot		
Properties	Property	Value
	Type	Reference

	Origin	AUTOSAR_ECUC
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3.2.3 EthTrcvConfig

Variable	EthTrcvConfig	
Description	Configuration of the individual transceiver.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.1 EthTrcvActAsSlavePassiveEnabled

Variable	EthTrcvActAsSlavePassiveEnabled	
Description	Specifies if the ECU is acting as a passive communication slave on the corresponding ComM channel	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.3.2 EthTrcvConnNeg

Variable	EthTrcvConnNeg	
Description	Specifies the connection negotiation of the Ethernet transceiver link.	
Screenshot		
Properties	Property	Value
	Type	Enumeration
	Origin	AUTOSAR_ECUC
	Range	ETHTRCV_CONN_NEG_AUTO Automatic Negotiation ETHTRCV_CONN_NEG_MASTER Master ETHTRCV_CONN_NEG_NONE PLCA ETHTRCV_CONN_NEG_SLAVE Slave
	Default	ETHTRCV_CONN_NEG_AUTO

3.2.3.3 EthTrcvDuplexMode

Variable	EthTrcvDuplexMode	
Description	Specifies the duplex mode of the Ethernet transceiver link if Auto-Negotiation is disabled.	
Screenshot		
Properties	Property	Value
	Type	Enumeration
	Origin	AUTOSAR_ECUC
	Range	ETHTRCV_DUPLEX_MODE_FULL Full duplex.

		ETHTRCV_DUPLEX_MODE_HALF Half duplex.
	Default	ETHTRCV_DUPLEX_MODE_FULL

3.2.3.4 EthTrcvForceSleepEnabled

Variable	EthTrcvForceSleepEnabled	
Description	Specifies if the Ethernet hardware (PHY) support to go to sleep without the confirmation of the Ethernet hardware (PHY) of the linked Ethernet communication partner according to the OA TC10.	
Screenshot	  	
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.3.5 EthTrcvIdx

Container	EthTrcvIdx	
Description	Specifies the instance ID of the configured transceiver.	
Screenshot	  0	
Properties	Property	Value
	Type	Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	0

3.2.3.6 EthTrcvMacLayerSpeed

Variable	EthTrcvMacLayerSpeed	
Description	Defines the baud rate of the MAC layer.	
Screenshot	  ETHTRCV_MAC_LAYER_SPEED_1G 	
Properties	Property	Value
	Type	Enumeration
	Origin	AUTOSAR_ECUC
	Range	ETHTRCV_MAC_LAYER_SPEED_100M, ETHTRCV_MAC_LAYER_SPEED_10G, ETHTRCV_MAC_LAYER_SPEED_10M ETHTRCV_MAC_LAYER_SPEED_1G ETHTRCV_MAC_LAYER_SPEED_2500M
	Default	ETHTRCV_MAC_LAYER_SPEED_1G

3.2.3.7 EthTrcvMacLayerSubType

Variable	EthTrcvMacLayerSubType	
Description	Defines the MAC layer subtype of a switch port	
Screenshot	  REDUCED 	
Properties	Property	Value
	Type	Enumeration
	Origin	AUTOSAR_ECUC

	Range	LIGHT,REDUCED, REVERSED, SERIAL, STANDARD, UNIVERSAL_SERIAL
	Default	REDUCED

3.2.3.8 EthTrcvMacLayerType

Variable	EthTrcvMacLayerType	
Description	Defines the MAC layer type of the ethernet transceiver.	
Screenshot		
Properties	Property	Value
	Type	Enumeration
	Origin	AUTOSAR_ECUC
	Range	ETH_MAC_LAYER_TYPE_XMII, ETH_MAC_LAYER_TYPE_XGMII, ETH_MAC_LAYER_TYPE_XXGMII
	Default	ETH_MAC_LAYER_TYPE_XGMII

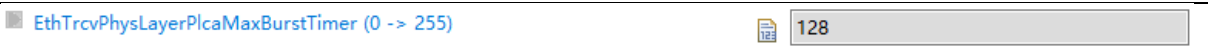
3.2.3.9 EthTrcvPhysLayerPlcaLocalNodeId

Variable	EthTrcvPhysLayerPlcaLocalNodeId	
Description	Configuration parameter for the transceiver node ID when the PLCA mode for 10BASE-T1S is used.	
Screenshot		
Properties	Property	Value
	Type	Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Range	0-255
	Default	255

3.2.3.10 EthTrcvPhysLayerPlcaMaxBurstCount

Variable	EthTrcvPhysLayerPlcaMaxBurstCount	
Description	Defines maximum packets allowed to be transmitted within a TO. The maximum burst count could differ per ECU within a PLCA mixed segment.	
Screenshot		
Properties	Property	Value
	Type	Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Range	0-255
	Default	255

3.2.3.11 EthTrcvPhysLayerPlcaMaxBurstTimer

Variable	EthTrcvPhysLayerPlcaMaxBurstTimer	
Description	Limits the burst frames in bit time. The maximum burst time could differ per ECU within a PLCA mixed segment.	
Screenshot		
Properties	Property	Value

	Type	Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Range	0-255
	Default	128

3.2.3.12 EthTrcvPhysLayerPlcaNodeCount

Variable	EthTrcvPhysLayerPlcaNodeCount	
Description	Defines the number of communication participants on the mixed segment. This value is relevant for the head-node transceiver (EthTrcvPhysLayerNodeID = 0) in order to know when the BEACON has to be transmitted.	
Screenshot		
Properties	Property	Value
	Type	Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Range	0-255
	Default	8

3.2.3.13 EthTrcvPhysLayerPlcaTransmitOpportunityTimer

Variable	EthTrcvPhysLayerPlcaTransmitOpportunityTimer	
Description	Timer for the transmission in bit time to evaluate if a Transmission Opportunity is yield or not.	
Screenshot		
Properties	Property	Value
	Type	Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Range	1-255
	Default	32

3.2.3.14 EthTrcvPhysLayerType

Variable	EthTrcvPhysLayerType	
Description	Specifies the physical layer type of the Ethernet transceiver link.	
Screenshot		
Properties	Property	Value
	Type	Enumeration
	Origin	AUTOSAR_ECUC
	Range	ETHTRCV_PHYS_LAYER_TYPE_1000BASE_T ETHTRCV_PHYS_LAYER_TYPE_1000BASE_T1 ETHTRCV_PHYS_LAYER_TYPE_100BASE_T1 ETHTRCV_PHYS_LAYER_TYPE_1000BASE_TX ETHTRCV_PHYS_LAYER_TYPE_10BASE_T1S
	Default	ETHTRCV_PHYS_LAYER_TYPE_1000BASE_T1

3.2.3.15 EthTrcvSleepModeExecutionDelay

Variable	EthTrcvSleepModeExecutionDelay	
Description	Specifies the time delay in seconds to execute a sleep (see OA TC10) for a Ethernet hardware (PHY), if a pending wake-up was detected while a ETH_MODE_DOWN was requested. The value shall be an integral multiple of EthTrcvMainFunctionPeriod.	
Screenshot		
Properties	Property	Value
	Type	Float
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Range	0-65.535
	Default	0.001

3.2.3.16 EthTrcvSleepRequestNumberOfRepetitions

Variable	EthTrcvSleepRequestNumberOfRepetitions	
Description	Specifies the repetitions to trigger a Sleep.Request (according to OA TC10) if a release of the communication channel was triggered by the upper layer (ETH_MODE_DOWN) and a SleepFail.Indication was signaled. Thus, the Ethernet hardware (PHY) of the connected communication partner did not accept the Sleep Request.	
Screenshot		
Properties	Property	Value
	Type	Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Range	0-255
	Default	0

3.2.3.17 EthTrcvSleepRequestRepetitionPeriod

Variable	EthTrcvSleepRequestRepetitionPeriod	
Description	Specifies the repetition period in seconds of repetitions for a Sleep Request (according to OA TC10). The value shall be a integral multiple of EthTrcvMainFunctionPeriod.	
Screenshot		
Properties	Property	Value
	Type	Float
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Range	0-65.535
	Default	0.001

3.2.3.18 EthTrcvSpeed

Variable	EthTrcvSpeed	
Description	Specifies the speed of the Ethernet transceiver link in [MBit/s]. If AutoNegotiation is enabled (EthTrcvConnNeg=TRCV_CONN_NEG_AUTO) this is the maximum speed advertised for Auto-Negotiation.	
Screenshot		

Properties	Property	Value
	Type	Enumeration
	Origin	AUTOSAR_ECUC
	Range	ETHTRCV_SPEED_10 ETHTRCV_SPEED_100 ETHTRCV_SPEED_1000
	Default	ETHTRCV_SPEED_1000

3.2.3.19 EthTrcvWakeUpCallout

Variable	EthTrcvWakeUpCallout	
Description	Configuration of the call-out name.	
Screenshot		
Properties	Property	Value
	Type	FunctionName
	Origin	AUTOSAR_ECUC
	Default	NULL_PTR

3.2.3.20 EthTrcvWakeupForwardLocalEnabled

Variable	EthTrcvWakeupForwardLocalEnabled	
Description	Specifies if remote wake up forwarding is enabled (TRUE) or disabled (FALSE) for OA TC10 compliant Ethernet Transceiver. If the parameter is set to TRUE, the Ethernet hardware (PHY) activate a local wake up (e.g. via I/O pin), if a remote wake-up	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.3.21 EthTrcvWakeupForwardRemoteEnabled

Variable	EthTrcvWakeupForwardRemoteEnabled	
Description	Specifies if local wake up forwarding is enabled (TRUE) or disabled (FALSE) for OA TC10 compliant Ethernet Transceiver.If the parameter is set to TRUE, the Ethernet hardware (PHY) transmit a wake-up (WUP or WUR on the data line (e.g. 100Base-T1), when a local wake-up occurred (e.g. via I/O pin), which was triggered by neighboring Ethernet hardware (PHY).	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.3.22 EthTrcvWakeupLocalDetectionTime

Variable	EthTrcvWakeupLocalDetectionTime
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Description	Defines the time in seconds when a local wake-up (e.g. via I/O pin) triggered by a neighboring PHY is evaluated as a valid wake-up.	
Screenshot		
Properties	Property	Value
	Type	Float
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Range	0.000001-0.001
	Default	0.001

3.2.3.23 EthTrcvWakeupLocalDurationTime

Variable	EthTrcvWakeupLocalDurationTime	
Description	Defines the duration time in seconds how long a local wake-up should be present on the local wake-up connection (e.g. via I/O pin) to indicate the neighboring PHYs about a wake-up.	
Screenshot		
Properties	Property	Value
	Type	Float
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Range	0.000001-0.001
	Default	0.001

3.2.3.24 EthTrcvWakeupLocalEnabled

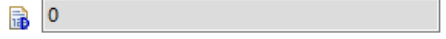
Variable	EthTrcvWakeupLocalEnabled	
Description	Specifies if local wake-up is enabled (TRUE) or disabled (FALSE) for OA TC10 compliant Ethernet hardware (PHY). If the parameter is set to TRUE, the Ethernet hardware (PHY) is able to detect and react on a wake-up received by a neighboring PHY (e.g. via I/O pin).	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.3.25 EthTrcvWakeupRemoteEnabled

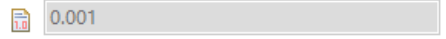
Variable	EthTrcvWakeupRemoteEnabled	
Description	Specifies if remote wake-up is enabled (TRUE) or disabled (FALSE) for OA TC10 compliant Ethernet hardware (PHY). If the parameter is set to TRUE, the Ethernet hardware (PHY) wake up when receiving a remote wake-up (e.g. via 100Base-T1 data line) triggered by the connected communication partner.	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False

	Default	False
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3.2.3.26 EthTrcvWakeupRequestNumberOfRepetitions

Variable	EthTrcvWakeupRequestNumberOfRepetitions	
Description	Specifies the repetitions to trigger a wake-up request (according to OA TC10), if an active communication request (ETH_MODE_ACTIVE_WITH_WAKEUP_REQUEST) was triggered by the upperlayer.	
Screenshot		
Properties	Property	Value
	Type	Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Range	0-255
	Default	0

3.2.3.27 EthTrcvWakeUpRequestRepetitionPeriod

Variable	EthTrcvWakeUpRequestRepetitionPeriod	
Description	Specifies the repetition period in seconds of a wake-up request, if an active communication request (ETH_MODE_ACTIVE_WITH_WAKEUP_REQUEST) was triggered by the upperlayer. The value shall be an integral multiple of EthTrcvMainFunctionPeriod.	
Screenshot		
Properties	Property	Value
	Type	Float
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Range	0-65.535
	Default	0.001

3.2.3.28 EthTrcvWakeupSleepOnDatalineEnabled

Variable	EthTrcvWakeupSleepOnDatalineEnabled	
Description	Specifies if wake-up on data line according to OA TC10 is supported by the used Ethernet hardware (PHY). If the parameter is set to TRUE, wake-up on data line is supported according to OA TC10.	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.3.29 EthTrcvConfigEcucPartitionRef

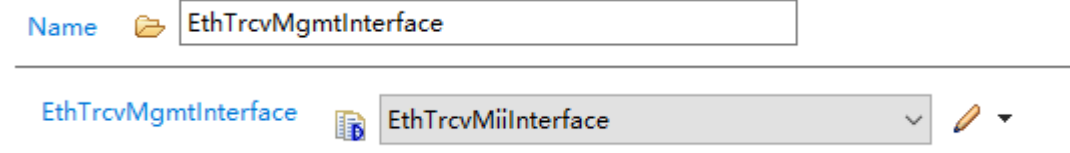
Variable	EthTrcvConfigEcucPartitionRef	
Description	Maps the Ethernet transceiver configuration to zero or one ECUC partitions. The ECUC partition referenced is a subset of the ECUC partitions where the Ethernet transceiver driver is mapped to.	
Screenshot		

Properties	Property	Value
	Type	Reference
	Origin	AUTOSAR_ECUC

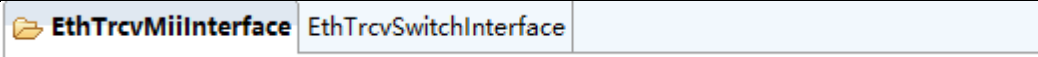
3.2.3.30 EthTrcvIcuChannelRef

Variable	EthTrcvIcuChannelRef	
Description	Reference to the IcuChannel to enable/disable the interrupts for wakeups.	
Screenshot		
Properties	Property	Value
	Type	Reference
	Origin	AUTOSAR_ECUC

3.2.3.31 EthTrcvMgmtInterface

Variable	EthTrcvMgmtInterface	
Description	The choice container allow to configure either the EthTrcv is accessed by a MII interface or Switch interface.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.31.1 EthTrcvMiInterface

Variable	EthTrcvMiInterface	
Description	This container includes the MII interface configuration between an Ethernet Controller and the Ethernet Transceiver. If this container is configured the EthTrcv shall call Eth_WriteMii / Eth_ReadMii API to access the hardware ethernet tranceiver.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.31.1.1 EthTrcvCtrlIdx

Container	EthTrcvCtrlIdx	
Description	Specifies the controller used for MII access to the transceiver	
Screenshot		
Properties	Property	Value
	Type	Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	0

3.2.3.31.1.2 EthTrcvMiIdx

Container	EthTrcvMiIdx
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Description	Specifies the transceiver index used for MII access to the transceiver.	
Screenshot		
Properties	Property	Value
	Type	Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	0

3.2.3.31.1.3 EthTrcvExtraMiildx

Container	EthTrcvExtraMiildx	
Description	Specifies the transceiver index used for MII access to the extra bridged transceiver.	
Screenshot		
Properties	Property	Value
	Type	Integer
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	0

3.2.3.31.1.4 EthDriverVendorIdAndApiInfix

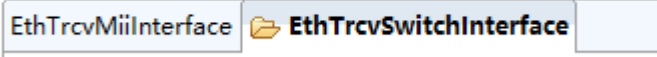
Variable	EthDriverVendorIdAndApiInfix	
Description	Set the vendor id and vendor api infix for the Eth driver. This is needed for EthTrcv driver to include the Eth header file and call mii indication functions.	
Screenshot		
Properties	Property	Value
	Type	String
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	NULL

3.2.3.31.1.5 EthTrcvMiiMuxCallout

Variable	EthTrcvMiiMuxCallout	
Description	Set the vendor id and vendor api infix for the EthTrcv driver. This is needed for Eth driver to include the transceiver header file and call mii indication functions.	
Screenshot		
Properties	Property	Value
	Type	Function Name
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	NULL_PTR

3.2.3.31.2 EthTrcvSwitchInterface

Variable	EthTrcvSwitchInterface	
Description	This container includes the Switch interface configuration between an Ethernet Switch and an Ethernet	

	Transceiver. If this container is configured the EthTrcv shall call EthSwt_WriteTrcvRegister / EthSwt_WriteTrcvRegister API to access the hardware ethernet transceiver.	
Screenshot		
Properties	Property	Value
	Type	Container

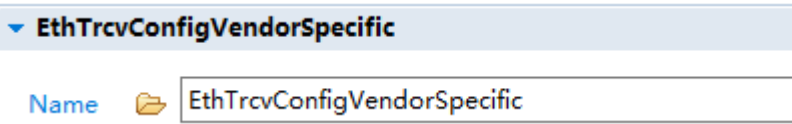
3.2.3.31.2.1 EthTrcvSwitchPortRef

Variable	EthTrcvSwitchPortRef	
Description	Symbolic name reference to [EthSwtPort]	
Screenshot		
Properties	Property	Value
	Type	Reference
	Origin	AUTOSAR_ECUC

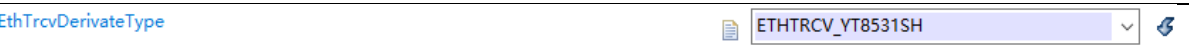
3.2.3.31.2.2 EthTrcvSwitchRef

Variable	EthTrcvSwitchRef	
Description	Reference to a switch configuration container.	
Screenshot		
Properties	Property	Value
	Type	Reference
	Origin	AUTOSAR_ECUC

3.2.3.32 EthTrcvConfigVendorSpecific

Variable	EthTrcvConfigVendorSpecific	
Description	Vendor specific configuration of the individual transceiver.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.32.1 EthTrcvDerivateType

Variable	EthTrcvDerivateType	
Description	Specifies the transceiver chip type.	
Screenshot		
Properties	Property	Value
	Type	Enumeration
	Origin	FLAGCHIP
	Range	ETHTRCV_YT8531SH: YT8531SH ETHTRCV_RTL8211FS: RTL8211FS ETHTRCV_RTL8211FS_RTL9010AA: RTL8211FS RTL9010AA bridge ETHTRCV_RTL8211FS_JL311X: RTL8211FS JL311X bridge

		ETHTRCV_YT8531SH_JL311X: YT8531SH JL311X bridge ETHTRCV_JL311X: JL311X ETHTRCV_JL311X_YT8531SH: JL311X YT8531SH bridge
	Default	ETHTRCV_YT8531SH_JL311X

3.2.3.32.2 EthTrcvSoftResetOnInit

Variable	EthTrcvSoftResetOnInit	
Description	Specifies if software reset is enabled (TRUE) or disabled (FALSE) on initialization.	
Screenshot	  	
Properties	Property	Value
	Type	Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	True

3.2.3.32.3 EthTrcvSoftResetOnInitRepetitions

Variable	EthTrcvSoftResetOnInitRepetitions	
Description	The repetitions to read software reset complete status after a software reset.	
Screenshot	  	
Properties	Property	Value
	Type	Integer
	Origin	FLAGCHIP
	Range	1 - 65535
	Default	1

3.2.3.32.4 EthTrcvRgmiiTxDelay

Variable	EthTrcvRgmiiTxDelay	
Description	Specifies the TX clock delay value in RGMII mode.	
Screenshot	  	
Properties	Property	Value
	Type	Integer
	Origin	FLAGCHIP
	Range	0 - 255
	Default	0

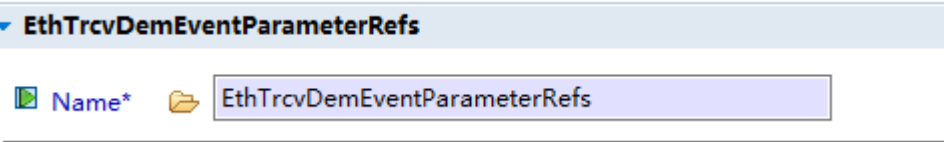
3.2.3.32.5 EthTrcvRgmiiRxDelay

Variable	EthTrcvRgmiiRxDelay	
Description	Specifies the RX clock delay value in RGMII mode.	
Screenshot	 	
Properties	Property	Value
	Type	Integer
	Origin	FLAGCHIP
	Range	0 - 255
	Default	0

3.2.3.32.6 EthTrcvOutPutClock

Variable	EthTrcvOutPutClock	
Description	Specifies the output clock mode.	
Screenshot		
Properties	Property	Value
	Type	Enumeration
	Origin	FLAGCHIP
	Range	ETHTRCV_CLK_OUTPUT_DISABLE, ETHTRCV_CLK_OUTPUT_25M ETHTRCV_CLK_OUTPUT_125M
	Default	ETHTRCV_CLK_OUTPUT_DISABLE

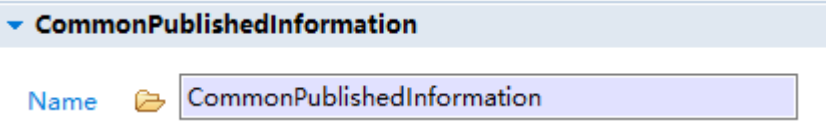
3.2.3.33 EthTrcvDemEventParameterRefs

Variable	EthTrcvDemEventParameterRefs	
Description	Container for the references to DemEventParameter elements which shall be invoked using the API Dem_SetEventStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId symbolic value. The standardized errors are provided in this container and can be extended by vendor-specific error references.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.33.1 ETHTRCV_E_ACCESS

Variable	ETHTRCV_E_ACCESS	
Description	Reference to the DemEventParameter which shall be issued when the error "Transceiver access failed" has occurred.	
Screenshot		
Properties	Property	Value
	Type	Reference

3.2.4 CommonPublishedInformation

Container	CommonPublishedInformation	
Description	Common container, aggregated by all modules. It contains published information about vendor and versions.	
Screenshot		
Properties	Property	Value
	Type	IDENTIFIABLE

3.2.4.2 AUTOSAR Major Version

Variable	AUTOSAR Major Version	
Description	Major version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	SynmbolicName	False
	Default	4
	Origin	FLAGCHIP

3.2.4.3 AUTOSAR Minor Version

Variable	AUTOSAR Minor Version	
Description	Minor version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	SynmbolicName	False
	Default	6
	Origin	FLAGCHIP

3.2.4.4 AUTOSAR Patch Version

Variable	AUTOSAR Patch Version	
Description	Revision version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	SynmbolicName	False
	Default	0
	Origin	FLAGCHIP

3.2.4.5 ModuleId

Variable	ModuleId
Description	Module ID of this module from Module List. Note: Implementation Specific Parameter

Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	SynmbolicName	False
	Default	73
	Origin	FLAGCHIP

3.2.4.6 Software Major Version

Variable	Software Major Version	
Description	Major version number of the vendor specific implementation of the module. The numbering is vendor specific. Note: Implementation Specific Parameter	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	SynmbolicName	False
	Default	0
	Origin	FLAGCHIP

3.2.4.7 Software Minor Version

Variable	Software Minor Version	
Description	Minor version number of the vendor specific implementation of the module. The numbering is vendor specific. Note: Implementation Specific Parameter	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	SynmbolicName	False
	Default	2
	Origin	FLAGCHIP

3.2.4.8 Software Patch Version

Variable	Software Patch Version	
Description	Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific. Note: Implementation Specific Parameter	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	SynmbolicName	False

	Default	0
	Origin	FLAGCHIP

3.2.4.9 VendorId

Variable	VendorId	
Description	Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list. Note: Implementation Specific Parameter	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	SynmbolicName	False
	Default	174
	Origin	FLAGCHIP

Chapter 4 Configuration Guides

4.1 Configuration Item Constraint

- At least one EthTrcvConfig should be configured for transceiver driver.

EthTrcv Usage Common Steps

- Users shall configure the general configurations for EthTrcv. This includes the EthTrcv functions enable and vendor specific configurations, etc.

The screenshot displays the configuration interface for the EthTrcv module, divided into two main sections: 'EthTrcvGeneral' and 'EthTrcvGeneralVendorSpecific'.

EthTrcvGeneral Section:

- Name:** EthTrcvGeneral
- Development Error Detection:** Enabled (checkbox checked)
- EthTrcvEnablePLCA:** Disabled (checkbox unchecked)
- EthTrcvGetBusWuReasonApi:** Disabled (checkbox unchecked)
- EthTrcvGetLinkStateApi:** Enabled (checkbox checked)
- EthTrcvGetTransceiverWakeupModeApi:** Disabled (checkbox unchecked)
- EthTrcvIndex (0 -> 255):** 0
- EthTrcvMainFunctionPeriod (0.001 -> 65.535):** 0.001
- EthTrcvMaxTrcvsSupported (1 -> 255):** 1
- EthTrcvSetPhyTestModeApi:** Disabled (checkbox unchecked)
- EthTrcvVersionInfoApi:** Enabled (checkbox checked)
- EthTrcvWakeUpSupport:** ETHTRCV_WAKEUP_NOT_SUPPORTED
- EthTrcvEnableCableDiagnosticApi:** Disabled (checkbox unchecked)
- EthTrcvGetBaudRateApi:** Enabled (checkbox checked)
- EthTrcvGetDuplexModeApi:** Enabled (checkbox checked)
- EthTrcvGetPhyIdentifierApi:** Enabled (checkbox checked)
- EthTrcvStartAutoNegotiationApi:** Enabled (checkbox checked)
- EthTrcvVersionInfoApiMacro:** Disabled (checkbox unchecked)

EthTrcvGeneralVendorSpecific Section:

- Name:** EthTrcvGeneralVendorSpecific
- EthTrcv Disable Diagnostic Event Module:** Disabled (checkbox unchecked)
- EthTrcvEnableUserModeSupport:** Disabled (checkbox unchecked)
- EthTrcvMulticoreSupport:** Enabled (checkbox checked)

- If EthTrcvMulticoreSupport is enabled, users shall configure the ECUC partition maps.

EthTrcvEcucPartitionRef
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EthTrcvEcucPartitionRef
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- 3) Users shall create at least one item in EthTrcvConfig. This container contains the configuration parameters of the ethernet transceiver(s). The EthTrcvConfig includes the autosar specified parameters and vendor specific parameters etc.

 EthTrcvActAsSlavePassiveEnabled	 ☐
EthTrcvConnNeg	 ETHTRCV_CONN_NEG_AUTO 
EthTrcvDuplexMode	 ETHTRCV_DUPLEX_MODE_FULL 
EthTrcvForceSleepEnabled	 ☒
EthTrcvIdx (dynamic range)	 0 
 EthTrcvMacLayerSpeed	 ETHTRCV_MAC_LAYER_SPEED_1G 
 EthTrcvMacLayerSubType	 REDUCED 
 EthTrcvMacLayerType	 ETHTRCV_MAC_LAYER_TYPE_XGMII 
 EthTrcvPhysLayerPlcaLocalNodeId (0 -> 255)	 255
 EthTrcvPhysLayerPlcaMaxBurstCount (0 -> 255)	 255
 EthTrcvPhysLayerPlcaMaxBurstTimer (0 -> 255)	 128
 EthTrcvPhysLayerPlcaNodeCount (0 -> 255)	 8
 EthTrcvPhysLayerPlcaTransmitOpportunityTimer (1 -> 255)	 32
 EthTrcvPhysLayerType	 ETHTRCV_PHYS_LAYER_TYPE_1000BASE_T 
 EthTrcvSleepModeExecutionDelay (0.001 -> 65.535)	 0.001
EthTrcvSleepRequestNumberOfRepetitions (0 -> 255)	 0
EthTrcvSleepRequestRepetitionPeriod (0.001 -> 65.535)	 0.001
EthTrcvSpeed	 ETHTRCV_SPEED_1000 
 EthTrcvWakeUpCallout	 NULL_PTR
EthTrcvWakeUpForwardLocalEnabled	 ☐ EthTrcvWakeUpForwardRemoteEnabled  ☐
 EthTrcvWakeUpLocalDetectionTime (0.000001 -> 0.1)	 0.001
 EthTrcvWakeUpLocalDurationTime (0.000001 -> 0.1)	 0.001
EthTrcvWakeUpLocalEnabled	 ☐ EthTrcvWakeUpRemoteEnabled  ☐
EthTrcvWakeUpRequestNumberOfRepetitions (0 -> 255)	 0
 EthTrcvWakeUpRequestRepetitionPeriod (0.001 -> 65.535)	 0.001
EthTrcvWakeUpSleepOnDataLineEnabled	 ☐
 EthTrcvConfigEcuPartitionRef*	 /EcuC/EcuC/EcuPartitionCollection_0/EcuPartition_0
 EthTrcvIcuChannelRef	

Name  EthTrcvMgmtInterface

EthTrcvMgmtInterface  EthTrcvMiiInterface 

EthTrcvMiiInterface	EthTrcvSwitchInterface
EthTrcvCtrlIdx (0 -> 255)	 0 
EthTrcvMiiIdx (0 -> 255)	 7 
EthTrcvExtraMiiIdx (0 -> 255)	 1 
EthDriverVendorIdAndApiInfix	 
EthTrcvMiiMuxCallout	 MiiPinSelect 

EthTrcvConfigVendorSpecific

Name

EthTrcvConfigVendorSpecific

EthTrcvDerivateType

ETHTRCV_YT8531SH

EthTrcvSoftResetOnInit

☒

EthTrcvSoftResetOnInitRepetitions (1 -> 65535)

3000

EthTrcvAutoNegotiationRepetitions (1 -> 999999)

75000

EthTrcvRgmiiTxDelay (0 -> 255)

12

EthTrcvRgmiiRxDelay (0 -> 255)

0

EthTrcvOutPutClock

ETHTRCV_CLK_OUTPUT_DISABLE

EthTrcvDemEventParameterRefs

Name*

EthTrcvDemEventParameterRefs

ETHTRCV_E_ACCESS

- 4) When configuration completed, click the Generate Project to generate relevant code.

